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Smart-substrate: a novel structural design to avert residual stress accretion in directed energy deposition additive manufacturing

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ABSTRACT

Residual stresses, related distortions and cracks are detrimental in metallic Additive Manufacturing (AM). Previously developed stress-control strategies based on reducing thermal gradients hardly diminish the stress concentrations at the built basement and easily affect other physical phenomena involved in AM. To overcome this, a novel strategy, named as Smart-Substrate, consisting of optimising the inner structure and local stiffness of the substrate is proposed to avert stress accretion and related part deformations. To demonstrate its advantages, a coupled thermomechanical finite element model for AM, experimentally calibrated with in-situ temperature and displacement measurements, is employed to analyse the thermal and mechanical behaviour of three groups of different structures with increasing geometrical complexity (single-wall, rectangular and block parts) fabricated by Directed Energy Deposit (DED) on the standard and smart substrates, respectively. Through using Smart-Substrate, the generation of residual stresses, especially the stress concentrations at the bottom corner of DED-builds being highly sensitive to cracks, and the induced deflections, are fundamentally throttled, and contrariwise for the standard substrate. More importantly, the use of Smart-Substrate is almost without prejudice to the temperature field, metallurgy and resulting mechanical hardness. This provides a possibility for addressing different physical problems individually, enlarging the AM process window.

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Additive manufacturing; structural design; residual stresses; multi-physics; thermomechanical simulation

1. Introduction

Metallic Additive Manufacturing (AM) is an advanced industrial technology to layer-by-layer fabricate complex parts and repair large components through using a focused high-energy heat source (e.g. laser, electron beam or arc) [1]. To date, several metallic AM techniques, like Directed Energy Deposition (DED) and Powder Bed Fusion (PBF), have been developed to meet the need of manufacturing a variety of products in industry [2,3]. Nevertheless, the AM process always is accompanied by high temperature gradients around the Heat Affected Zone (HAZ) as the printing proceeds. Due to the strong mechanical constraints from the cold substrate or previously deposited layers, the heated metal deposition cannot expand freely. As a consequence, residual restress are induced. Furthermore, the large number of heating and cooling cycles involved in the AM process exacerbates the accretion of residual stresses in AM-parts [4,5]. Once the magnitude of these stresses reaches the yield strength of the printed materials, plastic deformations or even cracks develop to release them [6,7]. Such defects easily cause the failure of 3D printing and significantly deteriorate the mechanical properties and performance of AM-parts.

To avert stress accretion during the deposition process, many strategies based on the principle of lowering the Maximum Temperature Gradient (MTG) have been developed, such as optimising AM variables and scanning patterns, and preheating substrate before or during AM process [8–11]. Nevertheless, their effectiveness is fairly limited due to the use of a high-energy heat source in metallic AM, which directly causes high MTG. In detail, the optimisation of processing parameters and printing path certainly narrows the process window of metallic AM, especially considered concurrently defect avoidance and microstructure design. Although

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preheating substrate can lower MTG and induced residual stresses, a higher preheating temperature easily leads to remarkable heat accumulation, which may accelerate the formation of harmful phases and microstructural coarsening, deteriorating the mechanical properties of AM deposits [2,12]. In addition, the minimisation of residual stresses in AM-builds can also be achieved by adopting auxiliary equipment (e.g. rolling) [13] or by post-process heat and mechanical treatments (e.g. high-temperature annealing or laser shock peening) [14–16]. However, these substantially increase both manufacturing cost and time and fail to rectify distortions and cracks induced by large residual stresses before being released. Therefore, the critical objective of this work is to seek an optimal solution to prevent stress accretion and related part deformations during the AM printing process but without prejudice to other physical processes (e.g. thermal field and metallurgy).

Residual stresses and part distortions essentially are mechanical outcomes from the repeated heating–cooling cycles during AM. Hence, more attention needs to be paid to the structure and stiffness of the components involved in AM because they markedly influence the mechanical response of the built parts.

For clarity, the formation mechanism of residual stresses during AM is visualised using a typical bar model including a central bar and two lateral bars with the same lengths [17], corresponding to the HAZ and the substrate in AM, respectively, as shown in Figure 1(a). The yielding of the middle bar in the thermal cycle is determined by the yield temperature, $T_Y = \left(\frac{S_{dep}}{S_{sub}} + 1\right)\frac{\sigma_Y}{\alpha \cdot E}$, where the physical significance of the included terms is present in Figure 1(a) and more details can be found in [17]. When the central bar experiences thermal cycling, its mechanical response (e.g. stress evolution) closely depends on the maximum heating temperature, T_{max}, responsible for MTG. In detail, three cases are involved and shown in Figure 1(b): (i) When $T_{max} \leq T_Y$ (along loading path OAO), no residual stresses develop; (ii) When $T_Y < T_{max} < 2T_Y$ (along path OABC), the residual stresses increase to the magnitude of the line segment |OC|; and (iii) When $2T_Y \leq T_{max}$ (along path OABDEF), the residual stresses rise to the yield stress. From this bar model, two core factors responsible for the stress generations in AM can be summarised: (i) MTG and (ii) yield temperature, T_Y. In the AM community, the former is well known, but this is not the case with the latter, related to the mechanical constraints during thermal deformations. As mentioned above, reducing MTG is very challenging, but it is possible to minimise residual stresses by increasing T_Y through adjusting S_{dep} and S_{sub} (Figure 1(b)), which correspond to the sizes of HAZ and substrate in AM, respectively. Typically, S_{dep} is very small and almost fixed, while it is feasible to decrease S_{sub} by substrate design to elevate T_Y.

Recently, some researchers [18,19] attempted using hollow substrate structures with some strip holes to control residual stresses in DED-parts. The results showed that the residual stress can be minimised by lowering the substrate stiffness. Yet, using such hollow substrates probably causes overheating due to the poor heat dissipation through the hollow structures, and negatively impact the metallurgical evolution as well as the performance of the final part [20,21]. To address this challenge, this work is focused on designing an optimal internal structure for the substrate to avoid the accretion of both residual stresses and deflections as well as heat accumulation in AM-parts, as shown in Figure 1(c). When adopting thick

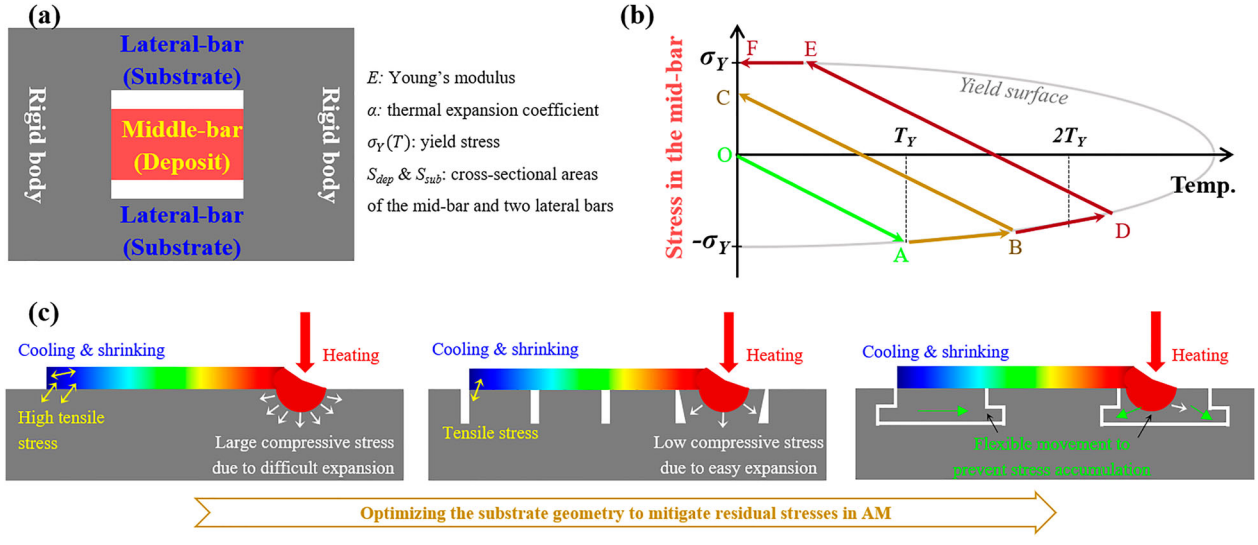


Figure 1. Residual stress formation in AM: (a) three bar model; (b) thermo-mechanical response of the central bar during thermal cycling; (c) residual stress control through the potential structural optimisation of the thick substrates.

substrates (e.g. > 10 mm) with a large horizontal cross-section, high tensile stresses and potential cracks are easily developed because of the strong mechanical constraint from the substrate even though it beneficially prevents part warpages [22–24]. Alternatively, if the local structural stiffness of the substrate is lowered by adding grooves or setting movable components in the base plate to reduce S_{sub} (see Figure 1(c)), T_Y is dramatically elevated. In this case, the stress accretion and the potential plastic strains transformed from the thermal deformations can be efficiently avoided because the metal deposition is allowed to freely expand/contract in an enlarged elastic range during thermal cycles in AM. Favourably, this small change in the structure of the substrate hardly affects the heat conduction from the build to the substrate and workbench. In this work, such modified substrate is named as 'Smart-Substrate', as it can automatically defuse thermal stresses induced by the AM process while cutting down the risk of overheating. Smart-substrate is intended as a new concept to enhance AM technology, particularly DED. The main advantage is the self-adjustable stiffness of the proposed substrate design to mitigate the residual stresses independently of AM-build. Thus, the proposed idea resembles the definition of smart materials [25,26].

To this end, a coupled thermo-mechanical Finite Element (FE) model for AM firstly is calibrated with in-situ experimental measurements from three groups of geometries: single-walls, rectangular and block parts fabricated by DED. Next, the validated model is used to study the influence of different substrate designs on the thermo-mechanical response. Finally, the advantage of the proposed Smart-Substrate approach is shown and its potential effect on the thermal evolution, metallurgy and mechanical hardness is also discussed.

2. Materials and experimental procedure

2.1. Laser-based DED process

Three sets of samples consisting of three single-walls, two rectangular structures and two blocks are additively manufactured by DED technique on seven annealed Ti6Al4 V substrates with a thickness of 11 mm, respectively, as shown in Figure 2. Noticeably, the design concept for the substrate structure is based on the thermo-mechanics in AM, which has been illustrated based on Figure 1 above. For the single-wall structures, three different substrates are used: a standard plate, and two plates with vertical grooves and with moveable trapezoidal tenons, corresponding to Wall-1, Wall-2 and Wall-3, respectively. For the rectangular geometries, two types of substrates are included: the standard one for Rectangle-1 and the plate with two moveable trapezoidal chunks for Rectangle-2. Similarly, a standard substrate and a modified plate assembled by several small tenons are used to build the Block-1 and Block-2 DED parts, respectively (see Figure 2). In detail, Figure 3 displays the design of the Smart-Substrate for building the block parts. Different from the standard substrate, the Smart-Substrate consists of two parts: (i) the basic structural framework and (ii) 16 separate and movable tenons inserted in the substrate framework. It can be seen from Figure 3(a) that the AM deposition region is composed of one central square table surrounded by 8 movable tenons; thereby, only the central small table may provide strong mechanical constraint to the AM-build, while the neighbouring tenons can flexibly adjust in the designed tip clearance to the expansion or shrinkage of the deposited layers during the AM process.

All the cutting patterns within the modified substrates are carried out by the electric discharge machining with a wire diameter of 0.25 mm. In addition, the movement of the tenons inserted in the substrate, driven by the thermal deformations induced by the DED process, just happens in the designed tip clearance (e.g. 0.25 mm).

The used DED machine is equipped with an *IPG YLS-10000-S2 T* ytterbium fibre laser with a wavelength of 1070 ± 5 nm and a maximum power of 10 kW, a five-axis numerical control workbench, a DPSF-2 high-accuracy adjustable automatic power feeder system, and an argon purged fabricating chamber with very low oxygen content (<0.8%). The metal powder used in this work is spherical Ti6Al4 V powder with a diameter range of 53~325 μm and low oxygen content. The powder is produced by a plasma electrode process and dried in a vacuum oven at 130°C for 2 h before AM.

A reciprocating scan strategy is used to build the single-walls, while two kinds of scan patterns defined by 4 different sequences repeated every 4-layers are adopted for the fabrication of the rectangular and block parts, respectively, as shown in Figure 4. Table 1 lists the process parameters used for the DED process and the corresponding size of the deposited workpieces shown in Figure 2.

A chemical solution composed of 1 ml HF, 3 ml HNO_3 and 46 ml H_2O is used as etching agent and the observation of the deposited microstructures is performed by optical microscopy. In addition, the Vickers hardness of the as-built block samples is tested; fifteen different points are selected to measure in each building position. The hardness tests are carried out with a load of 500 g and a load time of 15 s.

2.2. In-situ thermomechanical measurements

To acquire the in-situ measurements in terms of temperatures and displacements during DED process, one

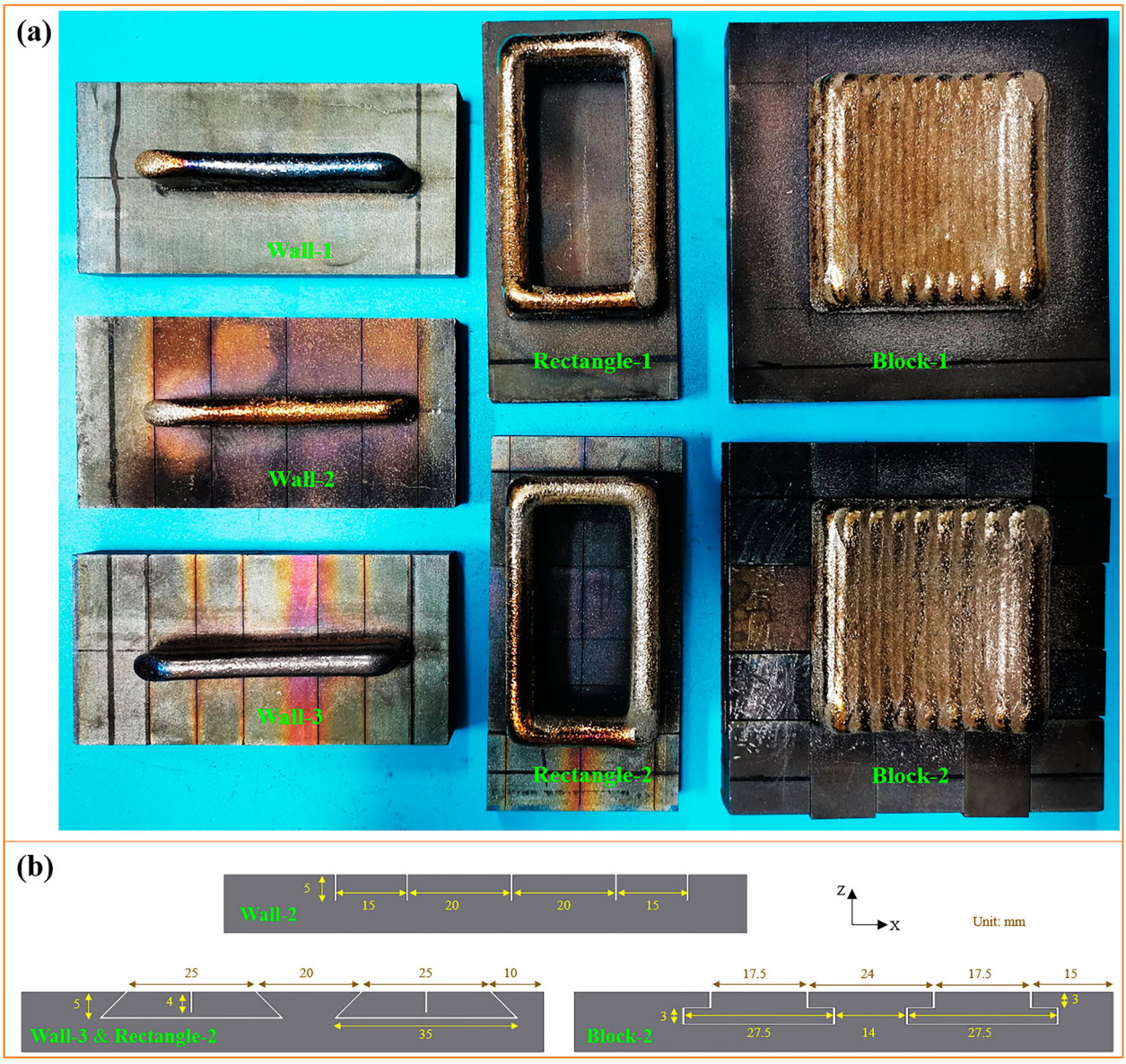


Figure 2. (a) Single-wall, rectangular and block parts printed on different substrates; (b) The sized details of the modified substrates with grooves or including several moveable components.

end of the substrate for all cases is clamped while the opposite end is allowed to freely deflect when new metal powder are deposited, as shown in Figure 5.

Figure 6 shows the dimension of the substrates and the locations of both the thermocouples (TC1∼TC9) and the displacement sensors (DS1∼DS3) to measure the temperature histories and the displacement evolutions of the lower surface of the substrates, respectively. Nine *Omega GG-K-30* type thermocouples with a measurement uncertainty of 2.2°C (±7.5%) and three *WXXY PM11-R1-20L* displacement sensors with a maximum range of 20.00 mm and an accuracy of 0.01% are adopted. A *Graphtec GL-900* high-speed data-logger with 8 receiver channels is employed to record all the displacement and temperature signals during the DED process.

3. FE modelling method

3.1. Thermomechanical simulation of DED

An in-house thermo-mechanical FE software, *COMET* [27,28], is employed to carry out the transient analysis during AM process. A time-marching scheme is adopted to advance in time and a staggered solution performs the thermal and mechanical analyses at each time-step, sequentially. Consequently, a coupled thermal-mechanical analysis is achieved including a transient heat transfer analysis of the thermal loading

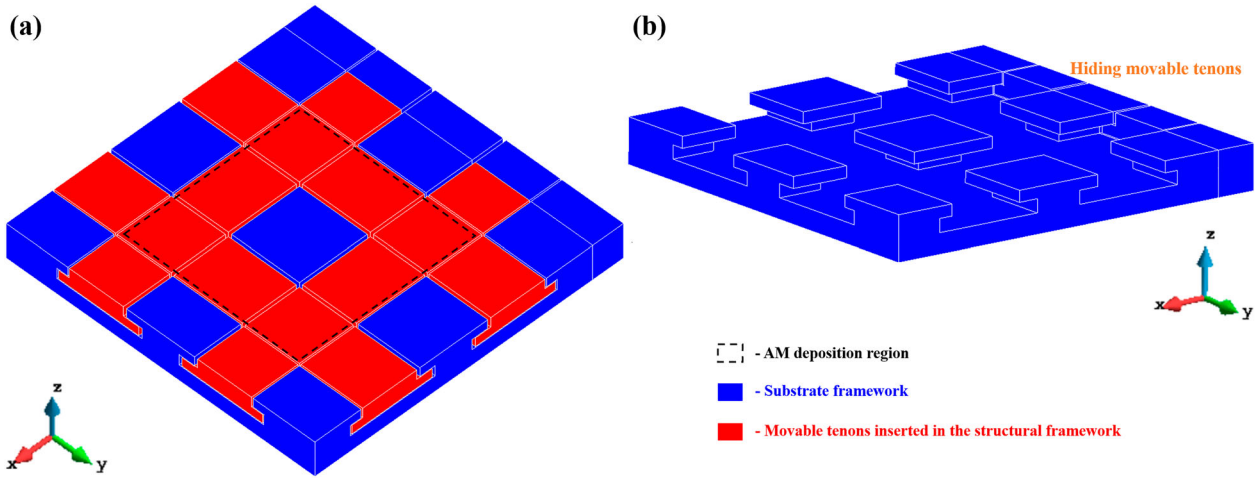


Figure 3. Smart-substrate for building the large-scale block part: (a) whole assembly structure; (b) basic structural framework.

induced by the DED process, followed by a stress analysis which is driven by the temperature field results from the thermal analysis. Because the laser energy input is much higher than the plastic dissipation, the thermo-mechanical problems are weakly coupled (single-way coupling). To simulate the DED process, the *birth–death-element* activation technique [29,30] is used to activate the elements belonging to each new deposition layer according to the user-defined building sequence.

In the simulation for DED, three assumptions are made: (i) A volumetric heat source is used because this study is focused on the macro-scale thermo-mechanical response rather than the melt flow in the molten-pool scale (small). At the global level, it is critical to properly consider the amount of energy introduced in the HAZ. (ii) The effect of the latent heat is negligible since the heat absorption during solid–liquid melting and the heat release during solidification are equal. (iii) The heat generated by plastic deformation is neglected in front of the laser input.

3.2. Governing equations

In the AM process, the heating and cooling are governed by the balance of energy equation:

$$\dot{H} = -\nabla \cdot q + \dot{Q} \tag{1}$$

where $\dot{H}$ is the enthalpy rate, and $\dot{Q}$ is the heat source, which is defined by the laser input power ($\dot{P}$), the molten-pool volume (V_{pool}), and the laser absorption efficiency (η):

$$\dot{Q} = \frac{\dot{P}\eta}{V_{pool}} \tag{2}$$

In this work, the molten-pool volume is defined as:

$$V_{pool} = D^2(T_l + h) \tag{3}$$

where D is the diameter of the laser beam, T_l and h are the layer thickness and the re-melting depth, respectively. For all simulations, $h = T_l$ and a uniform volumetric heat source is adopted, since the current research is

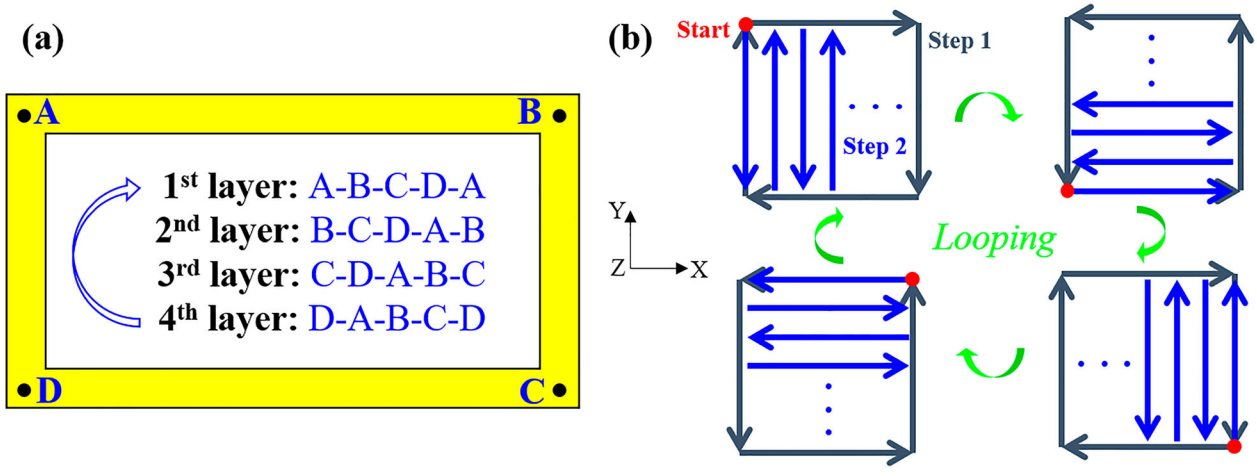


Figure 4. Scanning strategies used to print the (a) rectangular and (b) block parts.

Table 1. Process parameters utilised for DED and the size of different deposits.

	Laser power (W)	Scan speed (mm/s)	Up-lift height (mm)	Beam diameter (mm)	Hatch distance (mm)	Interval between adjacent layers (s)	Feeding rate (g/min)	Layer number (n)	Deposit sizes (mm3)
Wall-1	4000	20.0	0.4	5.0	-	0	15.0	48	70 × 7 × 17.5
Wall-2	4000	20.0	0.4	5.0	-	0	15.0	40	70 × 7 × 12
Wall-3	4000	20.0	0.4	5.0	-	0	15.0	48	70 × 7 × 17.5
Rectangle-1	4000	20.0	0.4	5.0	-	0	15.0	40	70 × 40 × 14
Rectangle-2	4000	20.0	0.4	5.0	-	0	15.0	40	70 × 40 × 14
Block-1	4000	20.0	0.6	5.0	3.0	200	15.0	20	60 × 60 × 11.6
Block-2	4000	20.0	0.6	5.0	3.0	200	15.0	20	60 × 60 × 13.2

focused on the macro thermal-mechanical response rather than the convective flow within the molten pool.

The heat flux q is expressed via Fourier's law:

$$q = -k\nabla T \tag{4}$$

where $k(T)$ and ∇T are the (temperature-dependent) thermal conductivity coefficient and the temperature gradient, respectively.

The heat loss due to convection is described via Newton's law:

$$q_{conv} = h_{conv}(T - T_{room}) \tag{5}$$

where h_{conv} is the Heat Transfer Coefficient (HTC) through convection, T is the surface temperature of the part and T_{room} corresponds to the ambient temperature.

The radiation heat flux is computed through Stefan-Boltzmann's law:

$$q_{rad} = \varepsilon_{rad}\sigma_{rad}(T^4 - T_{room}^4) \tag{6}$$

where ε_{rad} and σ_{rad} are the emissivity of the radiating surface and the Stefan–Boltzmann constant, respectively.

The stress analysis is performed in sequential thermal load steps adopting the temperature field obtained to solve the mechanical response. The governing equation for the stress analysis is the balance of momentum and the continuity equations written as:

$$\nabla \cdot s + \nabla p + b = 0 \tag{7}$$

$$(\nabla \cdot u - e^T) - \frac{p}{K} = 0 \tag{8}$$

where the Cauchy stress tensor $\boldsymbol{\sigma}$ is split into its spherical (pressure), p, and deviatoric, s, parts, respectively, as:

$$\boldsymbol{\sigma} = p\boldsymbol{I} + s(u) \tag{9}$$

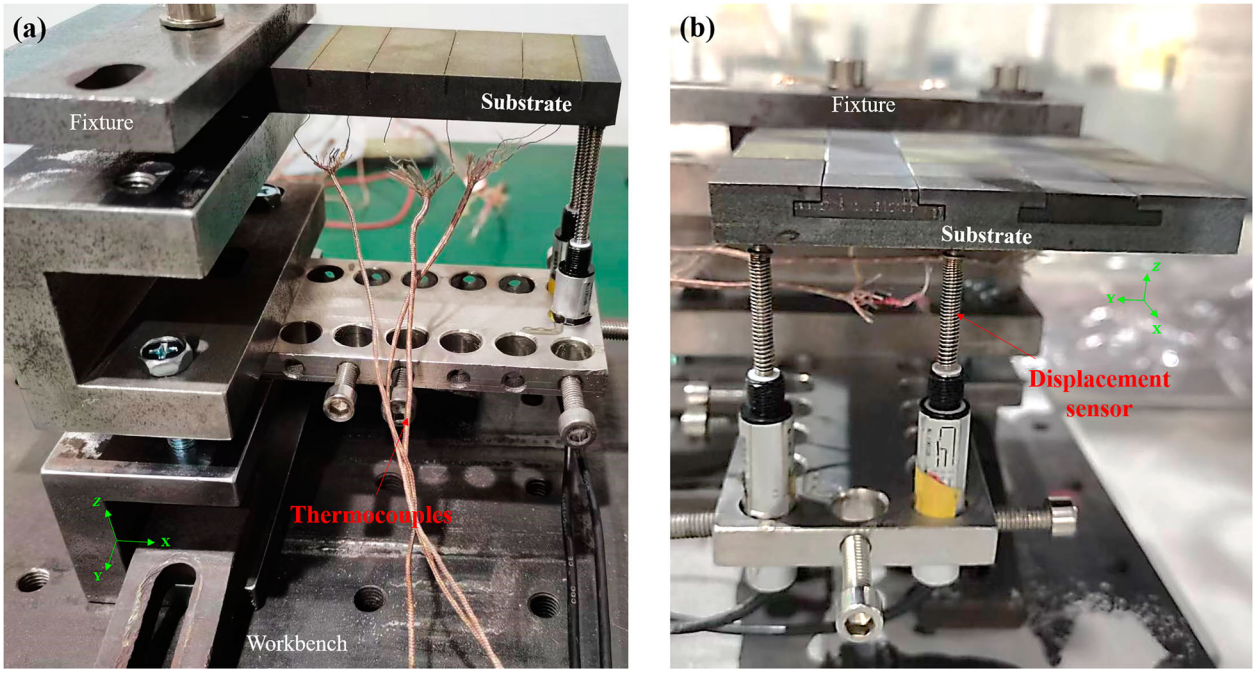


Figure 5. Experimental setup to monitor the temperature and vertical displacement of the bottom surface of the substrates used for building (a) single-walls and (b) block parts during DED.

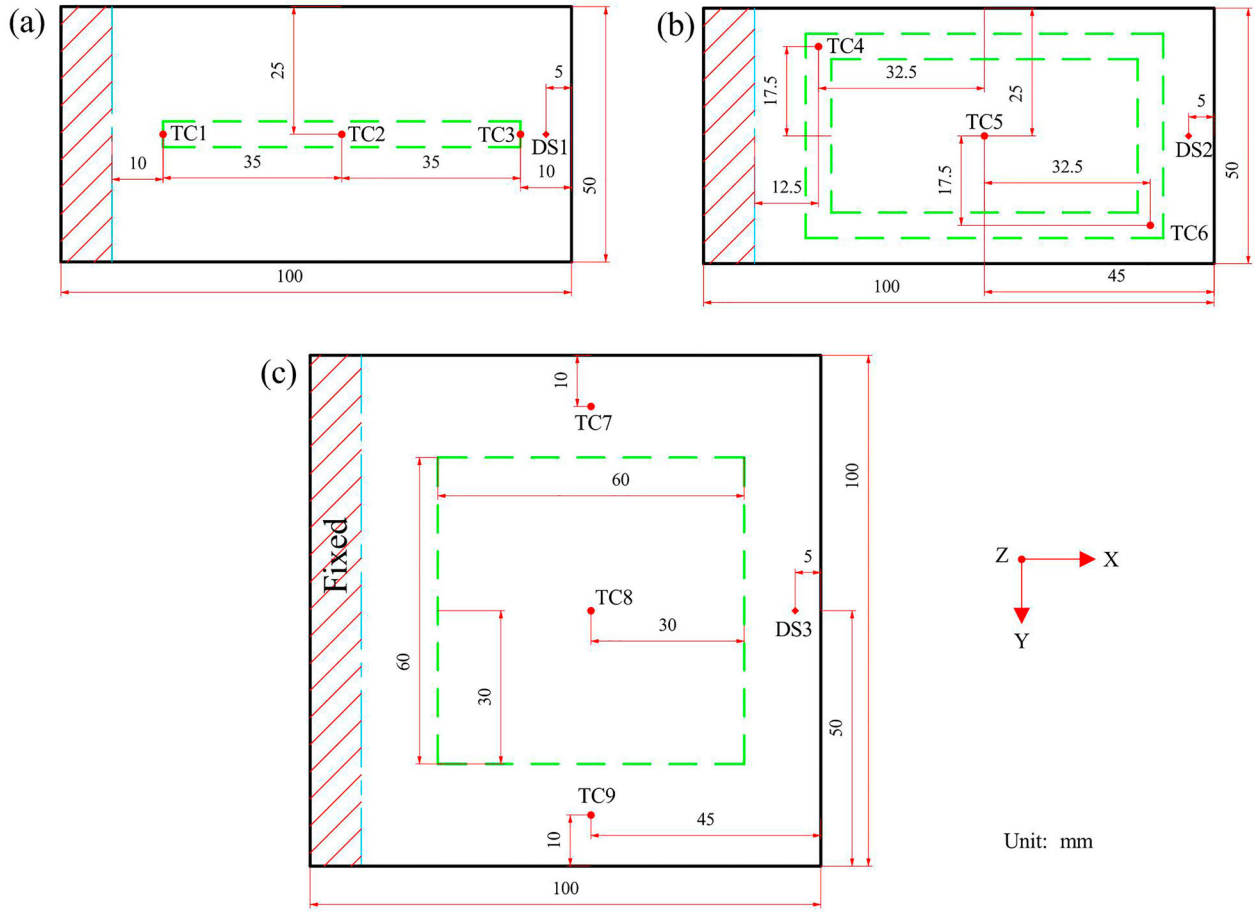


Figure 6. Substrate dimensions and the locations of the nine thermocouples (TC1~TC9) and the three displacement sensors (DS1~DS3) on the bottom surface of the substrates used for fabricating (a) single-wall, (b) rectangular and (c) block builds.

b denotes the body force (per unit of volume), and $K(T)$ are the (temperature-dependent) bulk modulus. The thermal deformation, e^T, is written as:

$$e^T(T, f_S) = e^{cool}(T) + e^{pc}(f_S) \tag{10}$$

where $e^{cool}(T)$ and $e^{pc}(f_S)$ are the thermal expansion and thermal contraction during the liquid/solid phase transformation, described as:

$$e^{cool}(T) = \alpha(T)(T - T_0) \tag{11}$$

$$e^{pc}(f_S) = \beta f_S \tag{12}$$

where α and β are the (temperature-dependent) coefficients of the thermal expansion and shrinkage, respectively, and T_0 and f_S are the initial temperature and the solid fraction, respectively.

In this work, a *J2*-thermo-elasto-visco-plastic model is used and the von-Mises yield-surface is defined as:

$$\Phi(s, q_h, T) = \|s\| - \sqrt{\frac{2}{3}}[\sigma_y - q_h] \tag{13}$$

where $\sigma_y(T)$ is the (temperature-dependent) yield stress allowing for the thermal softening while q_h is the stress-like variable controlling the isotropic strain-hardening.

The deviatoric counterpart of Cauchy's stress tensor is expressed as:

$$s = 2G(e - e^{vp}) \tag{14}$$

where G denotes the (temperature-dependent) shear modulus, while e and e^{vp} stand for the total (deviatoric) strain and the visco-plastic strain, respectively.

Ti6Al4 V titanium alloy is characterised by a solid-state phase transformation (SSPT), affecting the total strain. To consider the stress relaxation due to the alpha-beta phase transformation of Ti6Al4 V, Denlinger et al. [31] proposed using an annealing temperature of 690°C in the thermal-induced stress analysis of DED and achieved good agreement with the experimental data. Differently, Chen, Ye, and Xu [32] developed a thermo-mechanical model for wire-fed electron-beam freeform to investigate the SSPT temperature of Ti6Al4 V, and finally determined the annealing temperature as 850°C via variable temperature XRD measurements. The non-unique annealing temperature settings result from the difference in the definition of the

material constitutive laws used to simulate the mechanical behaviour of Ti6Al4 V in AM. For this reason, in-situ displacement measurement during DED is used in this work to seek the most suitable annealing temperature. In detail, the annealing temperature is set at $T_{ann} = 760^{\circ}C$ to consider the SSPT-induced stress relief. Once the temperature approaches T_{ann}, $\sigma_y \to 0$. Thereby, the deviatoric Cauchy stress reduces to:

$$s = \eta_{vp}\dot{e}^{vp} \tag{15}$$

where η_{vp} is the visco-plastic coefficient. Therefore, the material is featured by a purely viscous law when $T \geq T_{ann}$.

For a more detailed introduction of the governing equations and the constitutive laws implemented in this thermomechanical model, the reader may refer to previous works [33, 34].

3.3. Mesh model, material properties and boundary conditions

Figure 7 shows the FE meshes of seven different DED-components with increasing geometrical complexity from single-walls, rectangular parts to multi-pass multi-layer blocks. Table 2 presents the corresponding numbers of elements and nodes of all the geometries shown in Figure 7. The mesh size is set to $1.0 \times 1.0 \times t$ mm^3 (where t is equal to the layer thickness) for all the AM-parts based on a mesh convergence study [35], while coarser meshes are utilised for the substrates, balancing the prediction accuracy and the computational cost.

To ensure the computational accuracy of high-fidelity simulations, the step length is set as 4 mm (80% of the laser spot of 5 mm) and the time step size is $\Delta t = 0.2$ s. All the simulations are performed using a

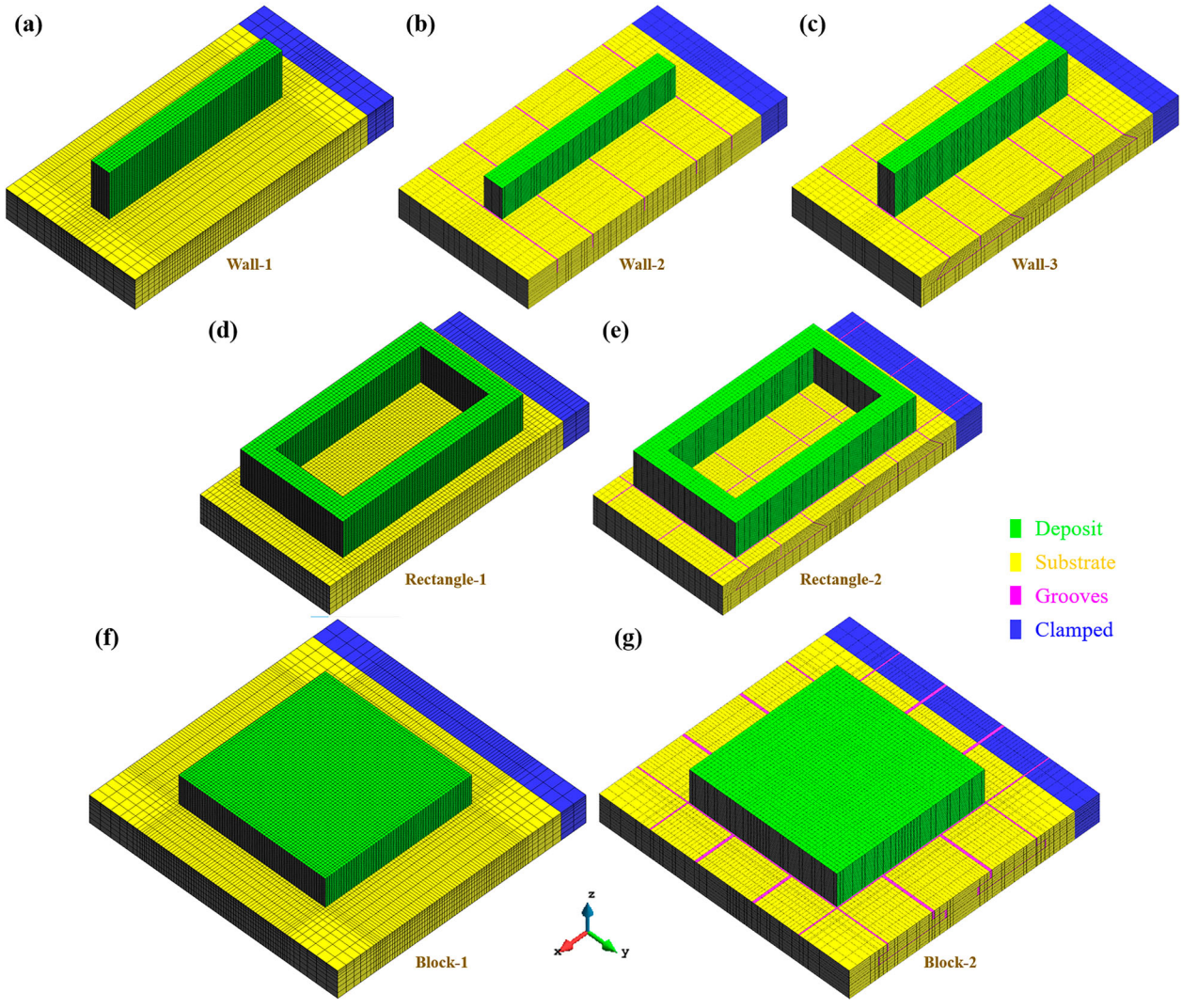


Figure 7. 3D FE meshes: (a) the standard substrate and the substrates (b) with vertical grooves and (c) with moveable trapezoidal tenons used for fabricating single-wall parts; (d) the standard substrate and (e) the substrate with moveable trapezoidal tenons for printing rectangular parts; (f) the standard substrate and (g) the substrate assembled by several small tenons for building block builds.

Table 2. Numbers of FE elements and nodes.

Samples	Number of hexahedral elements	Number of nodes
Wall-1	41,938	48,672
Wall-2	44,160	50,792
Wall-3	46,709	53,952
Rectangle-1	75,000	87,416
Rectangle-2	89,460	102,312
Block-1	116,992	126,395
Block-2	116,640	125,505

personal computer incorporating an Intel core i7-9700, 3.0 GHz processor and 16.0 GB of RAM. The total computational time takes 2.7, 14.2 and 150.4 h, corresponding to 433, 3045 and 16,777 steps, for the single-wall, rectangular and block parts, respectively.

Table 3 presents the temperature-dependent thermophysical properties of Ti6Al4 V titanium alloy [10] adopted for both the printed workpieces and the substrates in the computational simulations. Note that once the temperatures reach the melting point, the heat conductivity is increased to 83.5 W/(m·°C) to account for the convective flow inside the molten pool [36]. For the elements corresponding to the grooves (marked by fuchsia in Figure 6) within the modified substrates, the thermal and mechanical properties of Ti6Al4 V are set to 10% and 0.01% of the solids, respectively.

In the simulation of the DED process, the heat dissipation through both the convection and radiation mechanisms is considered for all the free surfaces of the built parts and the substrates. The calibrated emissivity and convective HTC are $\varepsilon_{rad} = 0.6$ and $h_{conv} = 10\ \mathrm{W/(m^2 \cdot {}^\circ C)}$, respectively. Also, the heat loss from the fixed surfaces of the substrate to the clamp system is approximated using an increased HTC of 100 W/(m^2·°C). The room temperature is assumed as $T_{room} = 25^\circ\mathrm{C}$ for all thermal analyses. The laser efficiency during DED is fixed at $\eta = 0.28$. An initial temperature of 800°C is set to the new elements in the deposition layers to account for the laser pre-heating to the fed powder before arriving at the molten pool.

In the mechanical simulations, the interfaces between the substrate and the fixture are fixed following the actual DED experimental setups. All the model parameters involved are obtained through comparing the numerical prediction with the in-situ thermo-mechanical measurements as presented in the following section.

4. Results and discussions

In this section, the thermally induced mechanical response of three types of printed geometries is analysed and discussed sequentially: (i) the simple single-wall structures; (ii) the complex hollow rectangular components; and (iii) the large-scale block parts. To achieve this, the coupled thermomechanical FE model applied in this research firstly is calibrated by comparing the in-situ measurement results of all the samples to guarantee the prediction accuracy, although it has been validated in previous works for optimising both the processing and material parameters [10,28]. Next, the mechanical results obtained from the dependable simulations are used to demonstrate the innovation and advantages of the proposed Smart-Substrate solution on controlling both the residual stresses and part warpages.

4.1. Single-wall components

Figure 8 compares the simulated and measured thermal histories and vertical displacements at different positions of the bottom of the substrate (Figure 6), used to fabricate the three single-wall parts. The agreement between the predicted curves (dash lines) and experimental evidence (solid lines) is noteworthy, especially for the temperature predictions. The small discrepancies are mainly attributed to the simplification of the boundary conditions (i.e. the effect of the clamp on the heat conduction is equivalent to an approximate HTC).

It can be seen from Figure 8(b,d,f) that during the printing process the substrate deflects downwards due to the thermal expansion of the deposited layers with a higher temperature than the base plate. Once the DED fabrication is finished, the free end of the substrate sharply warps upwards, especially for the cases of Wall-1

Table 3. Temperature-dependent material properties of Ti-6Al-4V.

Temperature (°C)	Density (kg/m^3)	Thermal Conductivity (W/(m·°C))	Heat Capacity (J/(kg·°C))	Poisson's Ratio	Thermal Expansion Coefficient (μm/(m·°C))	Young's Modulus (GPa)	Elastic Limit (MPa)
20	4420	7	546	0.345	8.78	110	850
205	4395	8.75	584	0.35	10	100	630
500	4350	12.6	651	0.37	11.2	76	470
995	4282	22.7	753	0.43	12.3	15	13
1100	4267	19.3	641	0.43	12.4	5	5
1200	4252	21	660	0.43	12.42	4	1
1600	4198	25.8	732	0.43	12.5	1	0.5
1650	3886	83.5	831	0.43	12.5	0.1	0.1
2000	3818	83.5	831	0.43	12.5	0.01	0.01

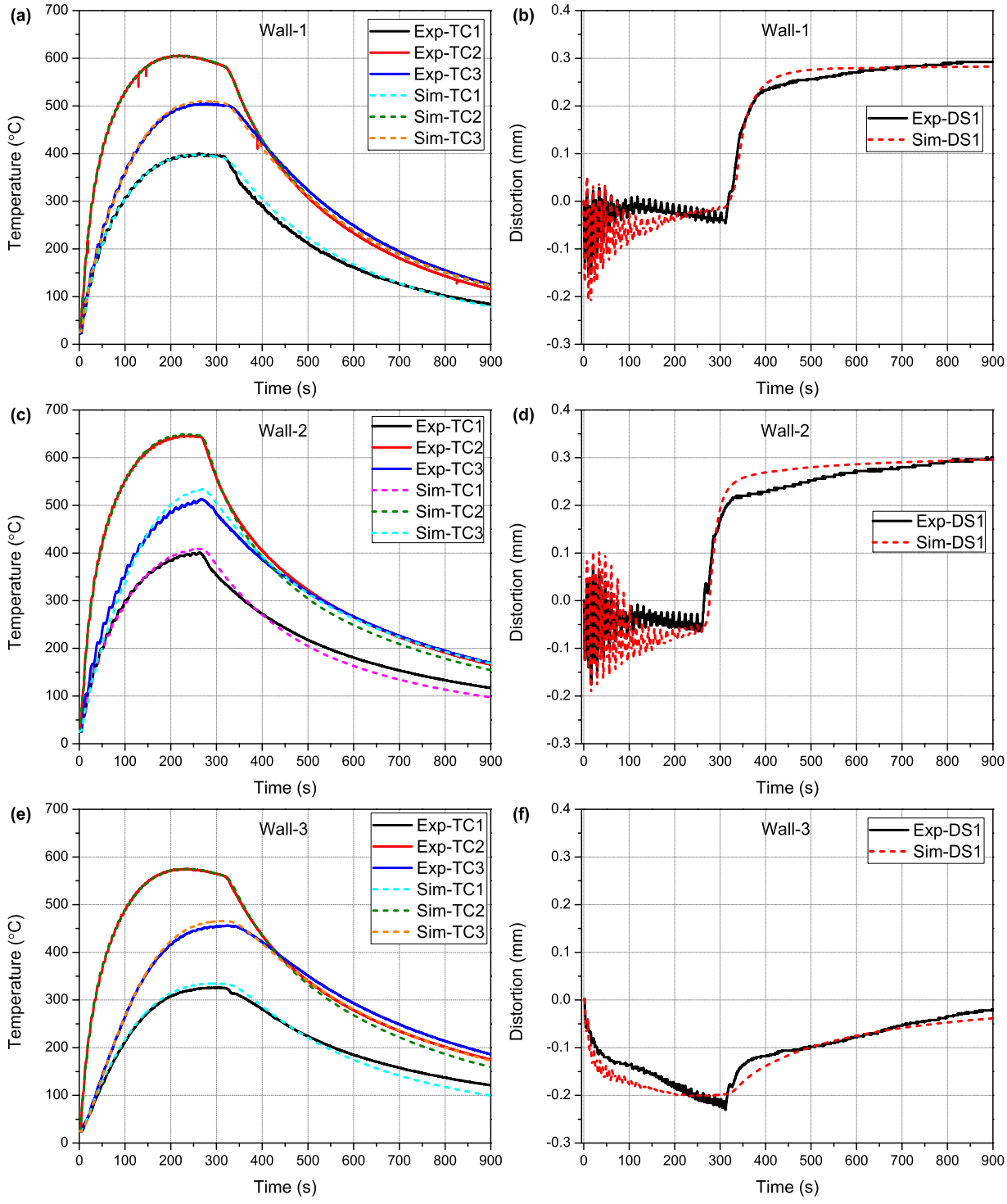


Figure 8. Single-walls: the evolutions of (a,c,e) the temperature histories and (b,d,f) the vertical displacement at the bottom surface of the substrates.

and Wall-2, because of the rapid cooling and contraction of the hot metal depositions. The comparison of the distortion evolution of three different single-wall samples illustrates that using the Smart-Substrate with an optimised inner structure significantly mitigates the final thermal deformations (only −0.02 mm) induced by the fabrication process (Figure 8(f)), compared to another two cases (up to 0.3 mm). These results are attributed to the evolution of the stress field, determined by the geometrical structures involved in the DED process.

Usually, the larger residual stresses are mainly in the direction of the scan line [37,38]; thereby, the contour-fills of the longitudinal stresses (σ_{xx}) and von Mises stresses of the three single-walls are computationally

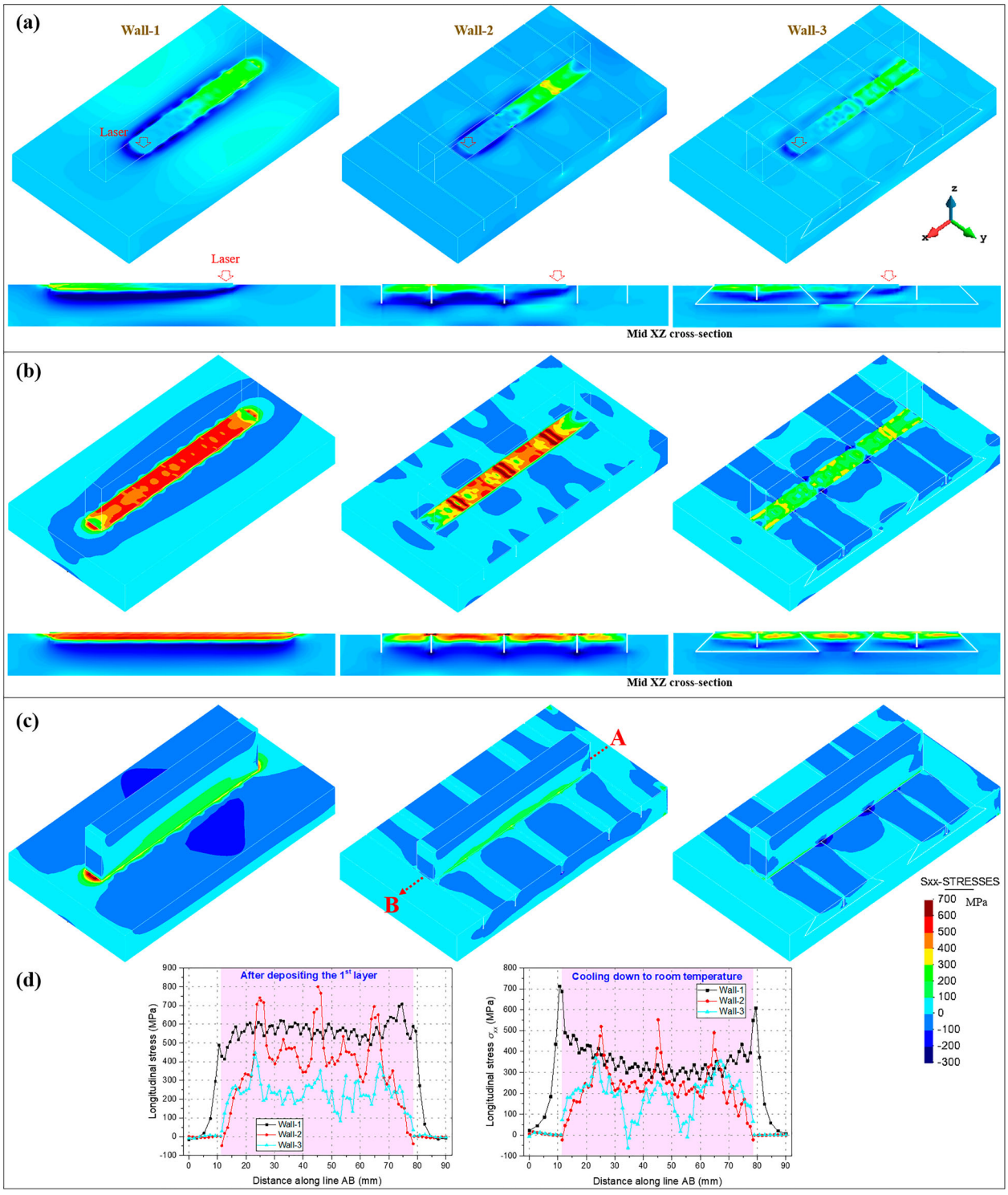


Figure 9. Single-walls: contour-fill of the longitudinal S_{xx} stresses (a) during the printing of the 1st layer; (b) after the deposition of the 1st layer; (c) after cooling down to room temperature; (d) the longitudinal stress distribution at the build-substrate surface (line AB).

simulated using the calibrated model and shown in Figures 9 and 10, respectively. It can be seen from Figure 9(a) and Figure 10(a) that during the printing of the 1st layer, a low stress is produced in the HAZ for the three substrate cases because the high-temperature molten pool softens the metal deposition, promoting stress relaxation through visco-plastic flow. As the heat source moves forward for the printing of the 1st layer, the metal depositions cool down and shrink quickly, leading to a tensile stress in the upper part of the substrate (corresponding to the past HAZ), while a compressive stress develops beneath the HAZ. Notably, the

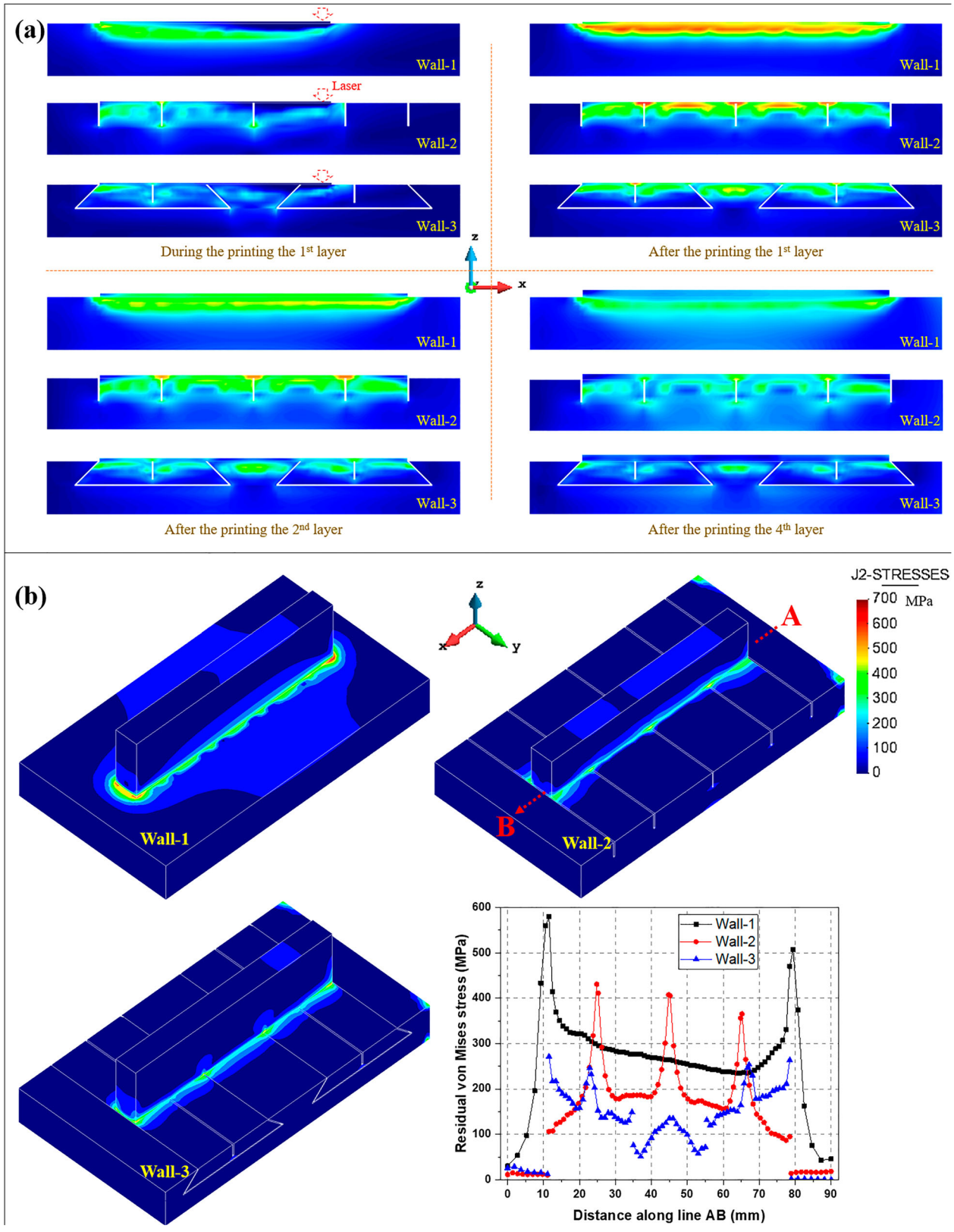


Figure 10. Single-walls: (a) evolution of the von Mises stress field at the mid XZ cross-section for different substrate samples; (b) contour-fill of the von Mises stresses and the stress distribution at the build-substrate surface.

magnitude of both tensile and compressive stresses in the Wall-1 case is pronouncedly higher than that in the other two samples, as shown in Figures 9(b, d) and 10 (a). Note that after building the 1st layer, the Wall-1 and Wall-2 samples show very high tensile stresses (up to 700 MPa). This is not the case for Wall-3 where the

stress level is much lower (about 350 MPa). The reason for this is that although the use of an intact substrate in the Wall-1 case ensures an overall structural stiffness, it also yields strong mechanical restraint to the deposited layer and, thus, high residual stresses. Differently, the substrate with vertical grooves in the Wall-2 case provides the clearance to allow the expanding of the heated material, reducing the development of plastic strains; also, it breaks the continuity of the stress distribution, reducing the stress accretion. The design of the moveable tenon inserted in the Wall-3 substrate further mitigates the generation of residual stresses, due to the significant freedom for thermal deformation during the DED process accompanying the repeated expanding/shrinking cycles.

It can be seen from Figure 10(a) that as the deposited layers increase, the stress level is lowered for all three cases. This is because the very short interval time (about 2 s) between adjacent layers and the high energy-input density (200 J/mm) allow the rapid accumulation of heat in the single-wall builds, functioning as in-situ heat treatment to release the residual stresses induced in the fabrication of the 1st layer. In addition, as the building wall grows, the lower printed material functions as a 'substrate' with an effectively reduced horizontal area (and reduced degree of mechanical constraint if compared to the original large substrate), reducing the formation of residual stresses. However, the magnitude of the residual stresses consistently ranks in the order of Wall-1 > Wall-2 > Wall-3. When the AM buildup is completed and cooling down to room temperature, a large tensile residual stress (700 MPa) is visible at the bottom end of the Wall-1 build, where cracks easily initiate and propagate, as shown in Figure 9(c,d) and Figure 10(b). Similar results are also reported in references [24,39–41]. Nevertheless, the Wall-2 and Wall-3 samples successfully prevent such stress concentrations (Figure 10(b)), especially for the latter (less than 280 MPa), because the tip clearance designed in the Smart-Substrate allows for a freer thermal expansion/contraction of materials during the AM process.

4.2. Rectangular components

Following the simple single-wall structures, in this section, the thermal and mechanical behaviour of the more complex rectangular parts is analysed. Figure 11 compares the calculated and experimental temperature and displacement histories at the lower surface of the substrates (see Figure 6) used to build the rectangular parts. The simulated results agree well with the experimental evidence.

The mechanical response of the rectangular parts is similar to the single-wall, as shown in Figure 11(b) and (d). In detail, the distortion evolution of the standard (Rectangle-1) and smart (Rectangle-2) substrates is different during the printing process. The displacement curve for the Rectangle-1 displays a more dramatic oscillation due to its stronger entire stiffness. However, both cases yield similar displacement values (about −0.22 mm) at the end of the deposition. Moreover, the Rectangle-1 case produces a sharp substrate bending upwards in the final cooling phase, leading to a remarkable residual deformation of approximate 0.28 mm, while a value of only 0.065 mm results for the Rectangle-2 case.

Noticeably, the deformation of the standard substrate for the Rectangle-1 case produces more dramatic fluctuations during the printing process as compared to the Wall-1 case (see Figure 8(b) and Figure10(b)). This is due to the existence of the four orthogonal thin walls in the rectangular structure, which yields a complex scanning process (Figure 4(a)) and, thus, a more complicated mechanical response during DED. For example, longitudinal bending of the substrate easily happens (deforms upwards) due to the contraction of the metal deposition when the printed long walls (along X-direction) cool down. To the contrary, the fabrication of the short-walls (along Y-direction) yields the transverse shrinkage of the upper part of the substrate, with a downward deflection. However, no obvious deformation fluctuation is found in both the Wall-3 and Rectangle-2 cases (see Figure 8(f) and Figure 11(d)) on account of the flexibility of the Smart-Substrate to avoid stress accretion.

Figures 12 and 13 show the evolution of von Mises stress field and the residual stresses in three orthogonal directions for both cases, respectively. In detail, Figure 12(a) shows the von Mises stress response at different scanning times when depositing the 1st layer. Note that when printing the second long track (corresponding to the scan-line 3) in the Rectangle-1 case, the first long track (scan-line 1) cools down and produces high tensile stress. At the same time, the printed first short track (scan-line 2) starts cooling, and high residual stresses are formed as the laser heat source moves forward to deposit the last short track (scan-line 4). When the 1st layer is completed, high residual stresses, more than 600 MPa, appear at the part-substrate interface region in the case of the standard substrate, similar to the Wall-1 case (see Figure 9(b)). Undoubtedly, such high stress level also is responsible for the large part deflection (Figure 11(b)). However, the Rectangle-2 case successfully avoids the generation of residual stresses in both the long and short tracks, even if their perpendicular geometrical relation increases the challenge of overcoming the stress problem.

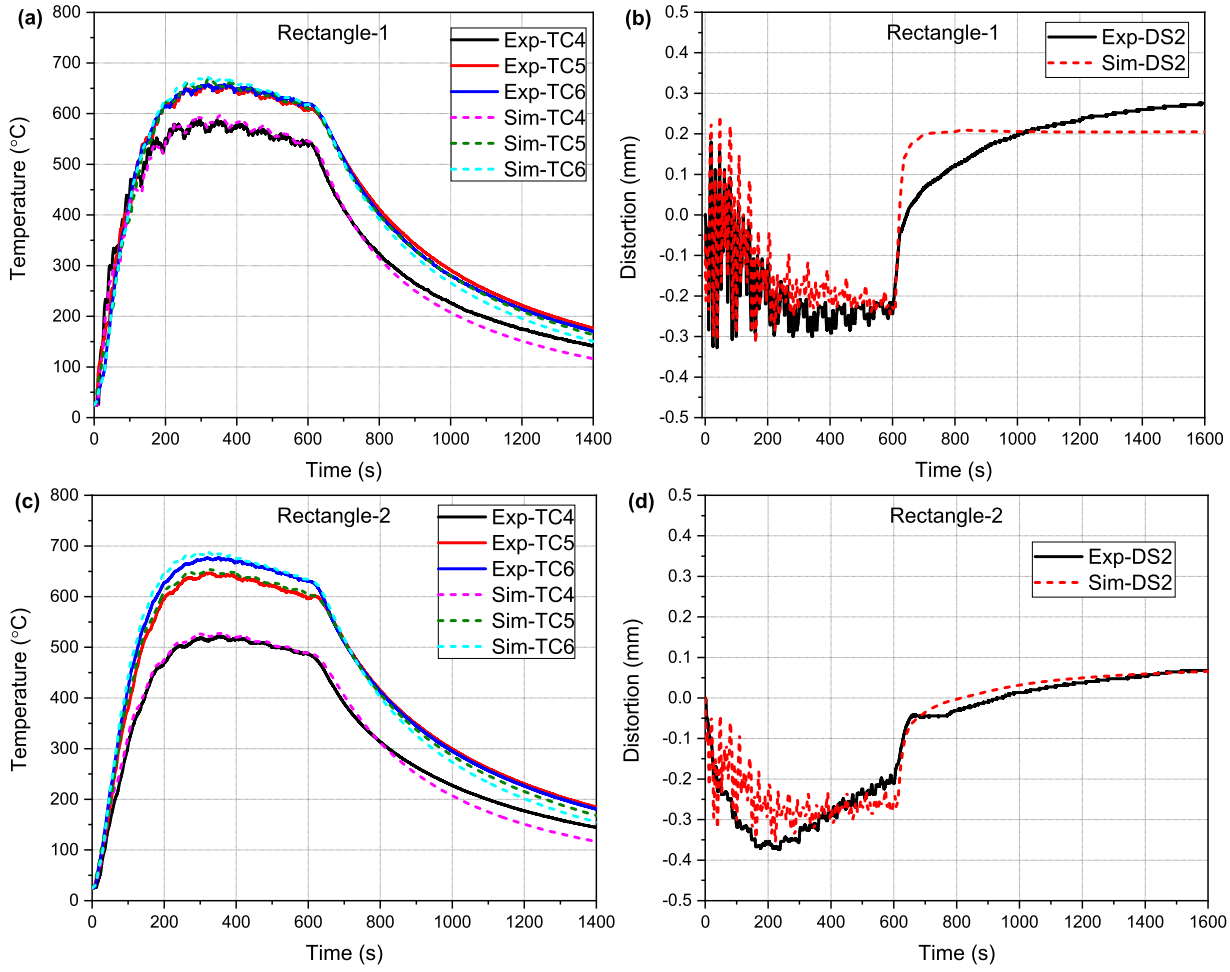


Figure 11. Rectangular part: the evolutions of (a,c) the temperature histories and (b,d) the vertical displacement at the free end of the substrates.

With the increase of the deposited layers, the standard and smart substrates still yield a high and lower stress level, respectively, as shown in Figure 12(b). This illustrates the effectiveness of using the Smart-Substrate on controlling residual stresses. Also, like in the single-wall parts, continuous printing causes an in-situ thermal annealing effect under high temperature (Figure 11), lowering the residual stresses in both the rectangular builds. Nevertheless, the stress concentrations at the bottom corners of the Rectangle-1 deposit are maintained until the build-up is finished, similar to the results reported by Yang et al. [42], while very lower stresses are accumulated in the Rectangle-2 case, as presented in Figures 12 and 13(c), thanks to the Smart-Substrate utilised in the deposition process, curbing the development of plastic deformations.

4.3. Block components

To ulteriorly verify the effectiveness of the proposed Smart-Substrate strategy on mitigating the residual stresses and deformations of industry-level AM components, two large-scale blocks are printed on the standard and smart substrates, respectively (Figure 2(a)).

As shown in Figure 14, the comparison of the numerically predicted and experimentally measured results in terms of the temperature and displacement histories at the bottom of the substrates for the two block samples, once again exemplifies the remarkable simulation precision of the thermo-mechanical model used.

It can be seen from Figure 14(b) that when the 1st layer of the Block-1 part is fabricated, the free end of the substrate displays a sharp increase in the vertical displacement, up to 1.1 mm (approximately equal to two layer-thicknesses), which is usually unacceptable during DED because it seriously impairs the geometrical precision of the built part. Furthermore, the displacement curve presents a pronounced fluctuation during the subsequent building process while the average value gradually increases to 1.64 mm until completing the whole build-up. Nevertheless, the final displacement is only 0.8 mm for the Block-2 sample (see Figure 14(d)),

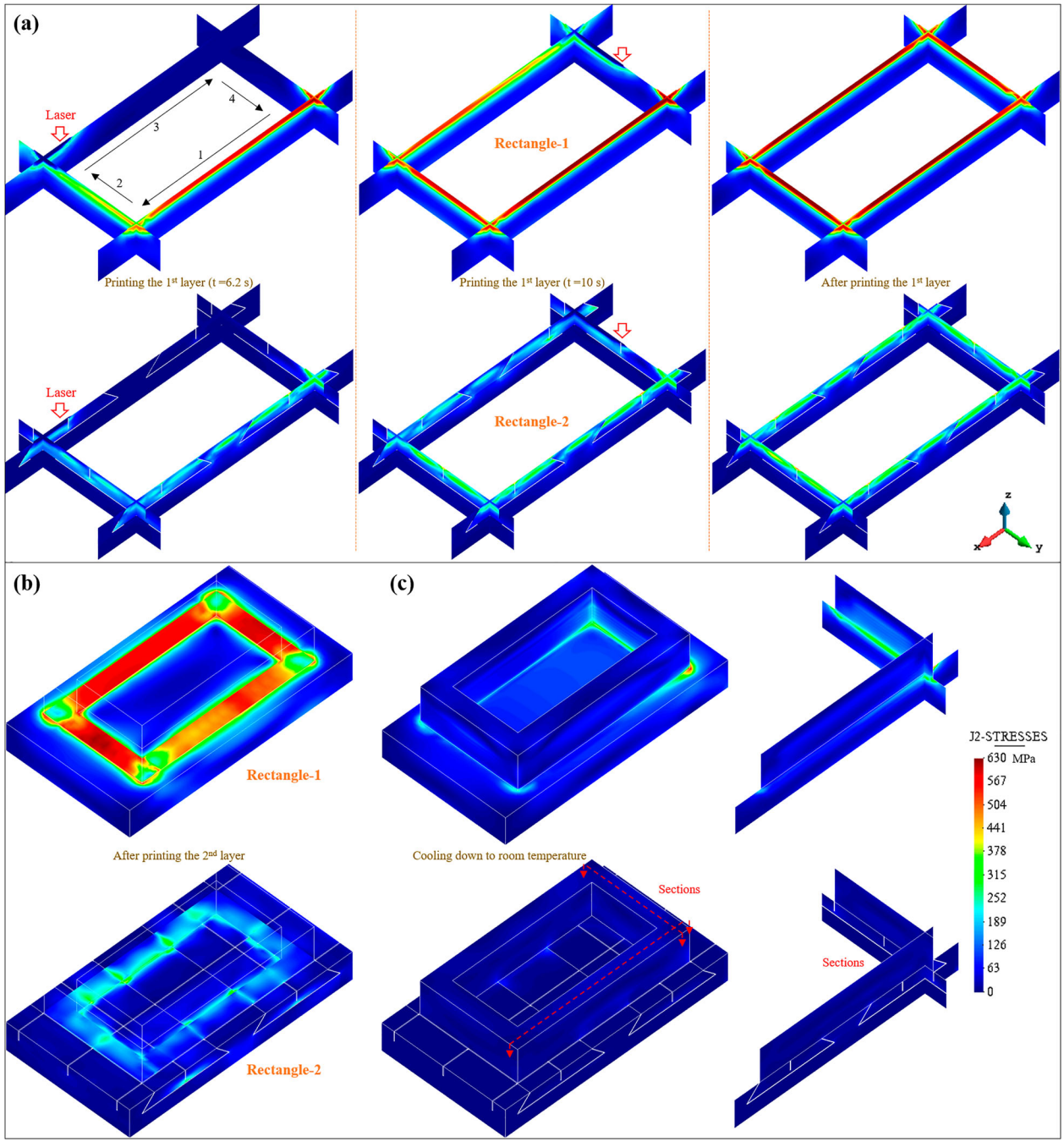


Figure 12. Rectangular parts: contour-fill of von Mises stress field: (a) cross-sections in the mid-line of the thin-walls at different printing times when depositing the 1st layer; (b) after the building of the 2nd layer; (c) cooling down to room temperature.

which is slowly accumulated during DED even if the structural stiffness of the Smart-Substrate is significantly weaker than the standard one.

Figures 15 and 16 show the evolution of von Mises stress field and the residual stresses in three orthogonal directions for the two block components, respectively. Observe that high residual stresses (up to 700 MPa) are induced after depositing the 1st layer for the standard case. However, unlike for the simple single-wall and rectangular samples (Figures 9 and 11(a)), the use of a 40% overrate (calculated from the beam diameter and hatch distance in Table 1) in the successive hatch-by-hatch scanning within one layer of the block samples yields smaller average stresses, since the tensile stresses formed in the current track are eliminated when depositing the following track. With the increase of the deposition layers, the thermally induced residual stresses gradually decrease due to the development of plastic deformation (Figure 14(b)) and the intrinsic high-temperature heat treatment (Figure 14(a)) facilitating stress

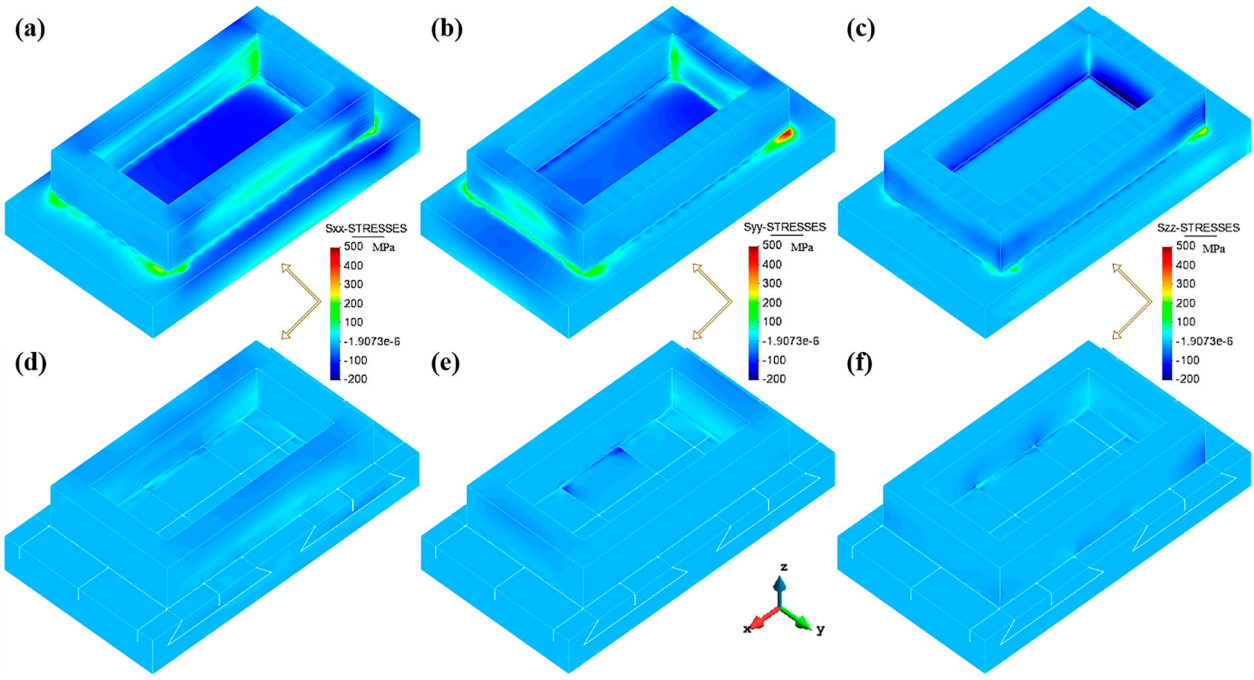


Figure 13. Rectangular parts: contour-fill of the residual stresses in three orthogonal directions (S_{xx}, S_{yy} and S_{zz} for X, Y and Z directions, respectively): (a–c) standard substrate; (d–f) Smart-Substrate.

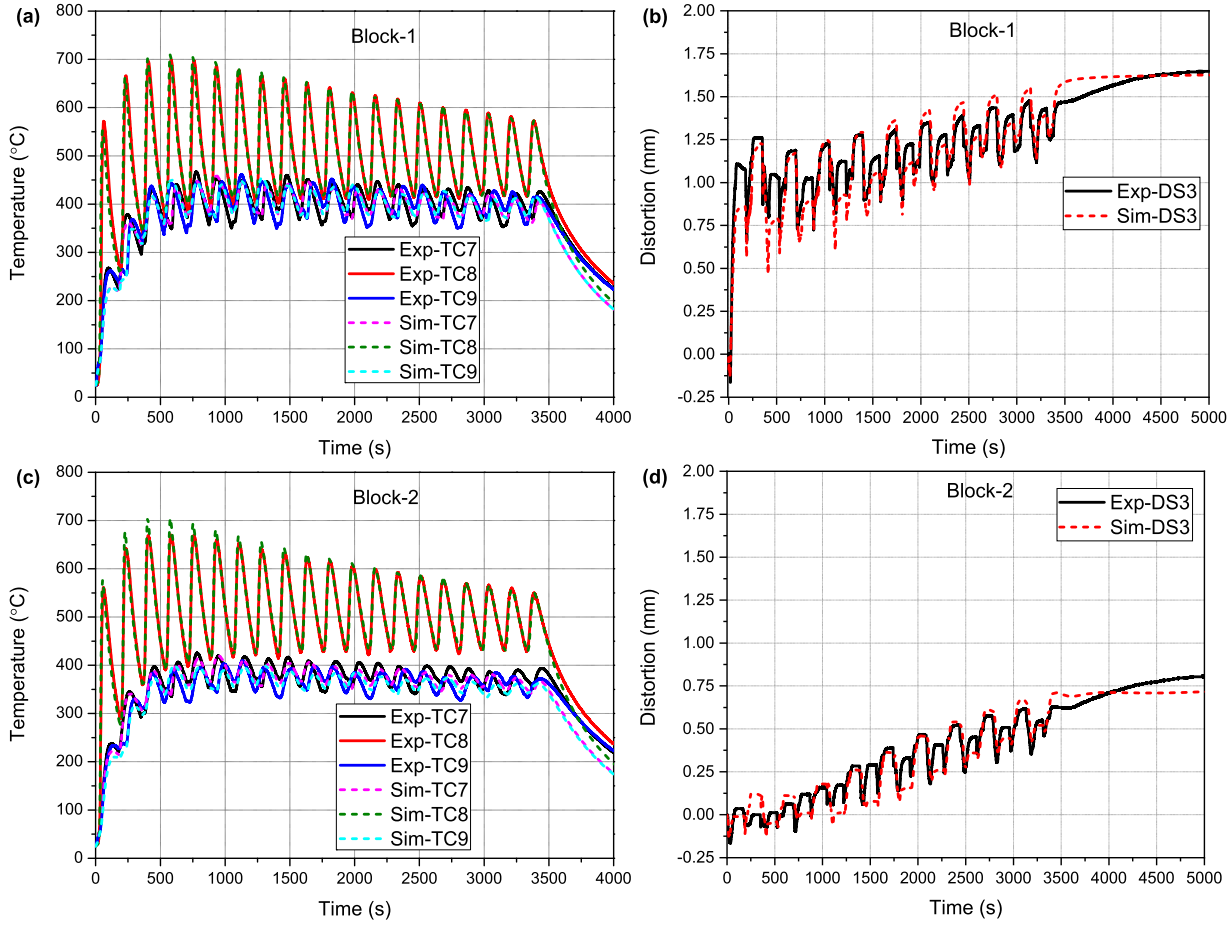


Figure 14. Block parts: the evolutions of (a,c) the temperature histories and (b,d) the vertical displacement at the free end of the substrates.

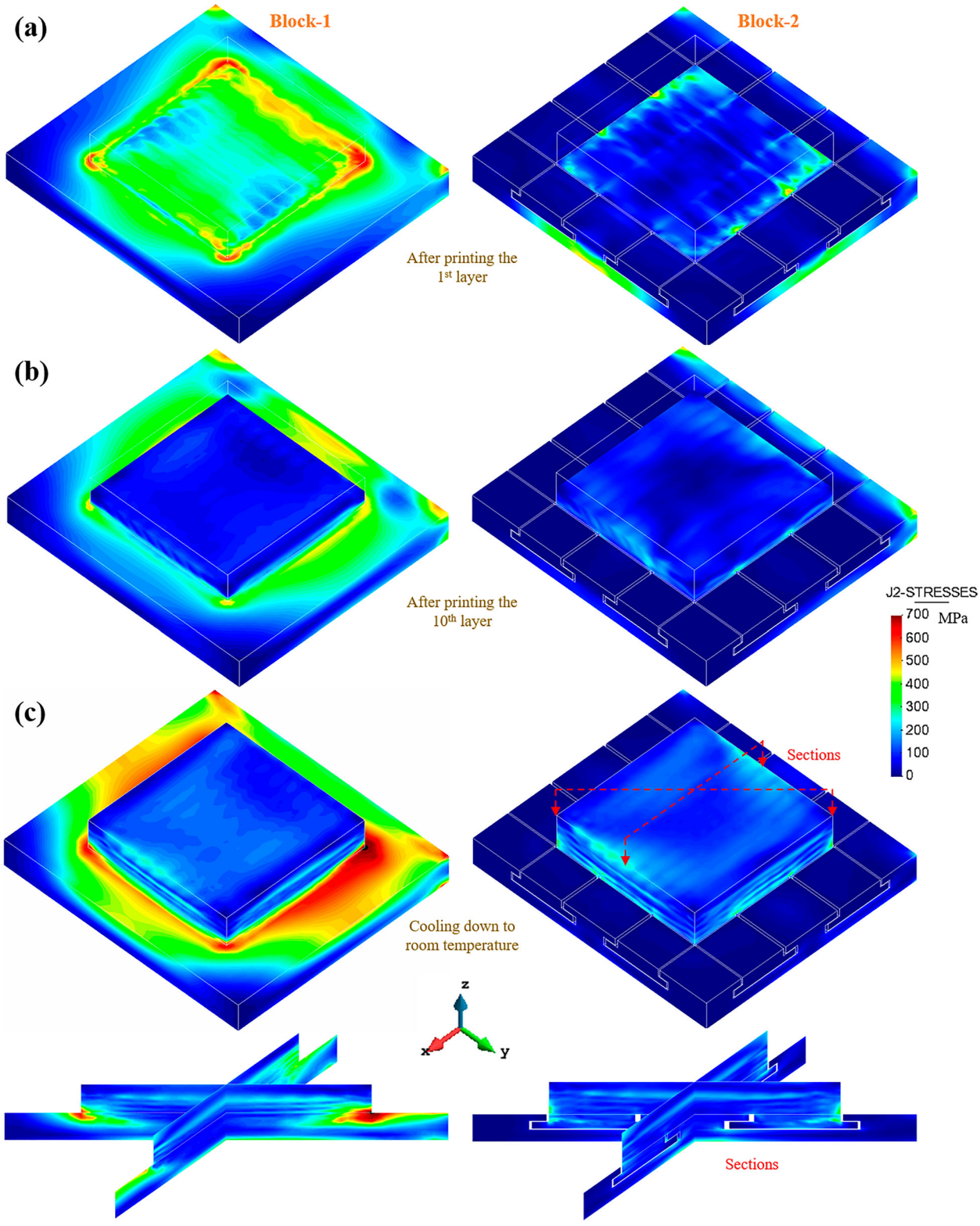


Figure 15. Block parts: the von Mises stress field after building the (a) 1st and (b) 10th layer; (c) cooling down to room temperature.

relief, but the stress concentrations still hold at the basement of the builds, especially in the corners (Figures 15 and 16). However, by using the protective Smart-Substrate strategy, the residual stresses can be effectively reduced. Thus, a lower stress level (less than 300 MPa) is achieved during the whole DED printing process, and distortion is also minimised by roughly 50% (Figure 14(d)). In addition, although the use of a long

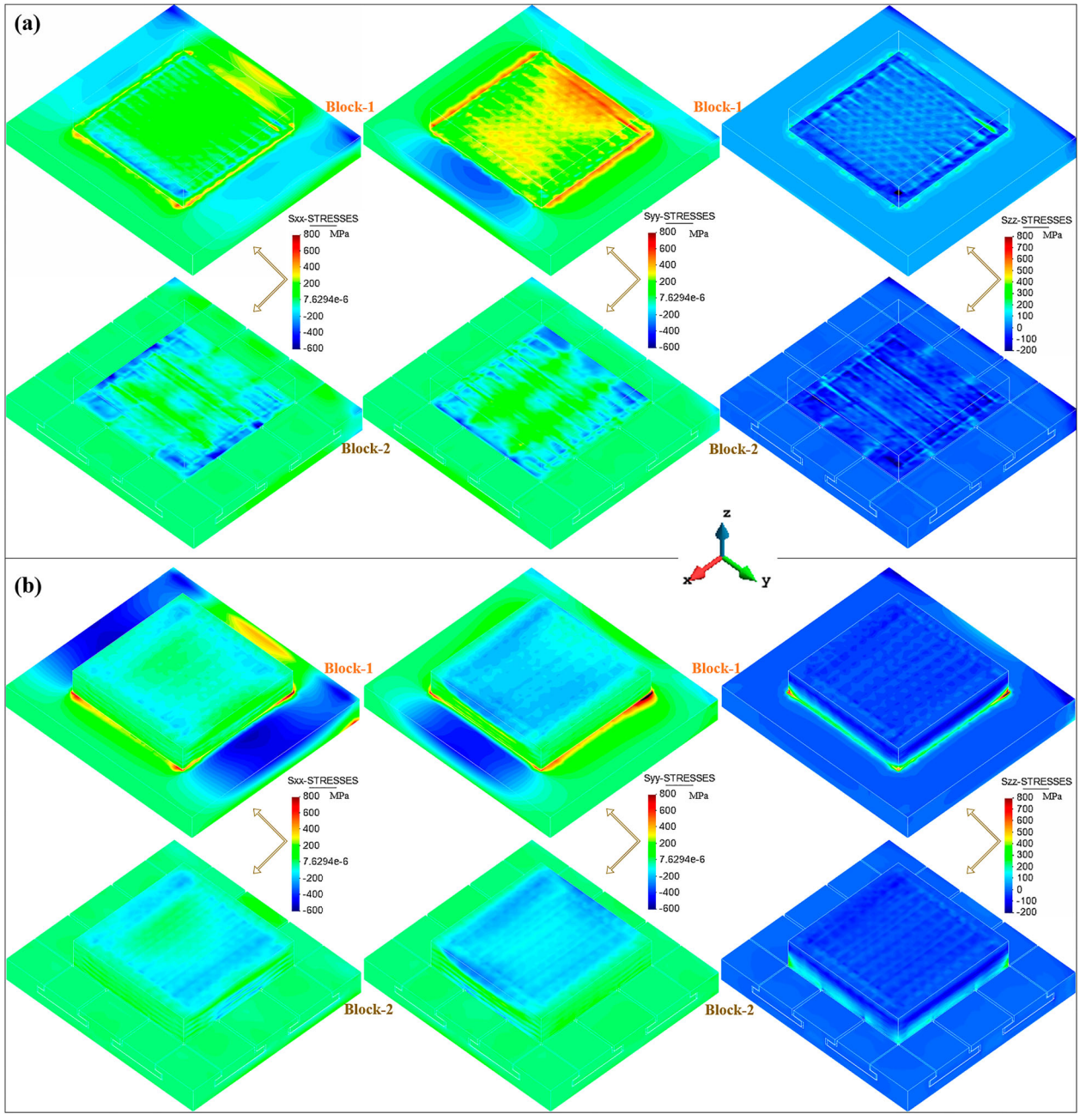


Figure 16. Block parts: the stress fields in three orthogonal directions (S_{xx}, S_{yy} and S_{zz} for X, Y and Z directions, respectively): (a) after the building of the 1st layer; (b) after the build-up and cooling down to room temperature.

dwell time of 200 s during the building the block parts favours stress accretion [31], this disadvantage is completely avoided by applying Smart-Substrate.

4.4. Influence of applying smart-substrate on other physical fields

As mentioned, most of in-situ stress-control strategies developed and based on mitigating MTG inevitably affect the thermal distribution, related metallurgical evolution and potential defect formation. Also, they fail to diminish the stress concentrations at the deposit-substrate interface, especially the corners [43–45], because where large MTG and strong mechanical constraints coexist. Differently, the Smart-Substrate strategy proposed is designed to increase the yield temperature (by lowering the local mechanical constraint of the substrate) instead of reducing MTG to avert the generation of residual stresses during DED process. As known, AM is a complex multi-physics multi-scale interaction including a large number of input parameters, non-uniform thermal field,

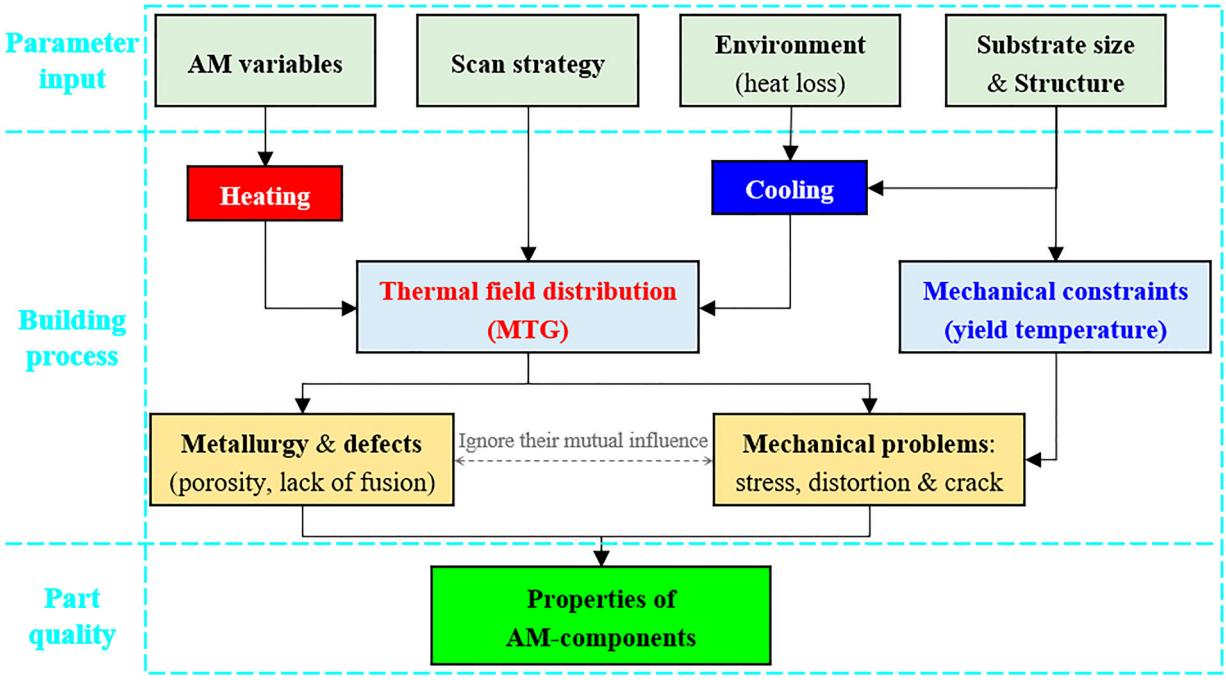


Figure 17. Complex interactions between different physical fields involved in AM.

mechanical response, metallurgical evolution and defect formation, all of which affect the final part properties, as shown in Figure 17. Thereby, the potential impact of using the Smart-Substrate approach to other physical processes must be assessed.

The thermal histories, the deposited macro/microstructures, the corresponding size of the β-grains and α laths, as well as the microhardness, at the centre and different heights of the block parts manufactured on the standard and smart substrates, respectively, are extracted and compared in Figures 18–21, respectively. Note that both cases present similar temperature histories and resulted macro/microstructures and microhardness for different building positions. This illustrates the independence of the Smart-Substrate strategy in controlling residual stresses and deflections during AM process without compromising other physical phenomena. This makes it unnecessary to consider the stress problem when optimally determining AM variables for achieving superior microstructure and avoiding defect formation. This is beneficial for widening the process window of metallic AM.

4.5. Advantages and limitations of the smart-substrate strategy

According to the stress results presented in this work (see Figures 9, 11, and 15) and the residual stress distribution in AM-parts reported in previous papers [23,35,46,47], it is found that large residual stresses (or stress concentrations) typically appear at the part-substrate interface, especially at the bottom end or corners of the build. Such phenomenon is explained as follows. When the first layers of the metallic material are deposited on the cold substrate, a large MTG and strong mechanical constraint from the substrate inevitably yield high residual stresses and related part deformations and/or cracks. As the building height increases, the high stresses yielded in the deposition of the first layers are slightly decreased due to the gradually increased heat accumulation (favouring stress annealing) and the weakened mechanical constraint provided by the previously deposited layers, but cannot be erased because of the use of stiff substrates. For this reason, the Smart-Substrate solution is based on the thermo-mechanics of AM and efficiently averts stress accretion by allowing the development of thermal deformations while preserving the overall structural stiffness of the substrate. Although the Smart-Substrate works at the built basement, it also guarantees a pronounced reduction (by more than 60%) of residual stresses in the whole build even with an increased height. This can be proven by comparing the predicted residual stress fields of two single-wall parts with an increased height of 50 mm fabricated on the standard and smart substrates respectively, as shown in Figure 22.

In addition to the typical rectangular structures presented in this work, many other components with curved shapes can also use the smart substrate to

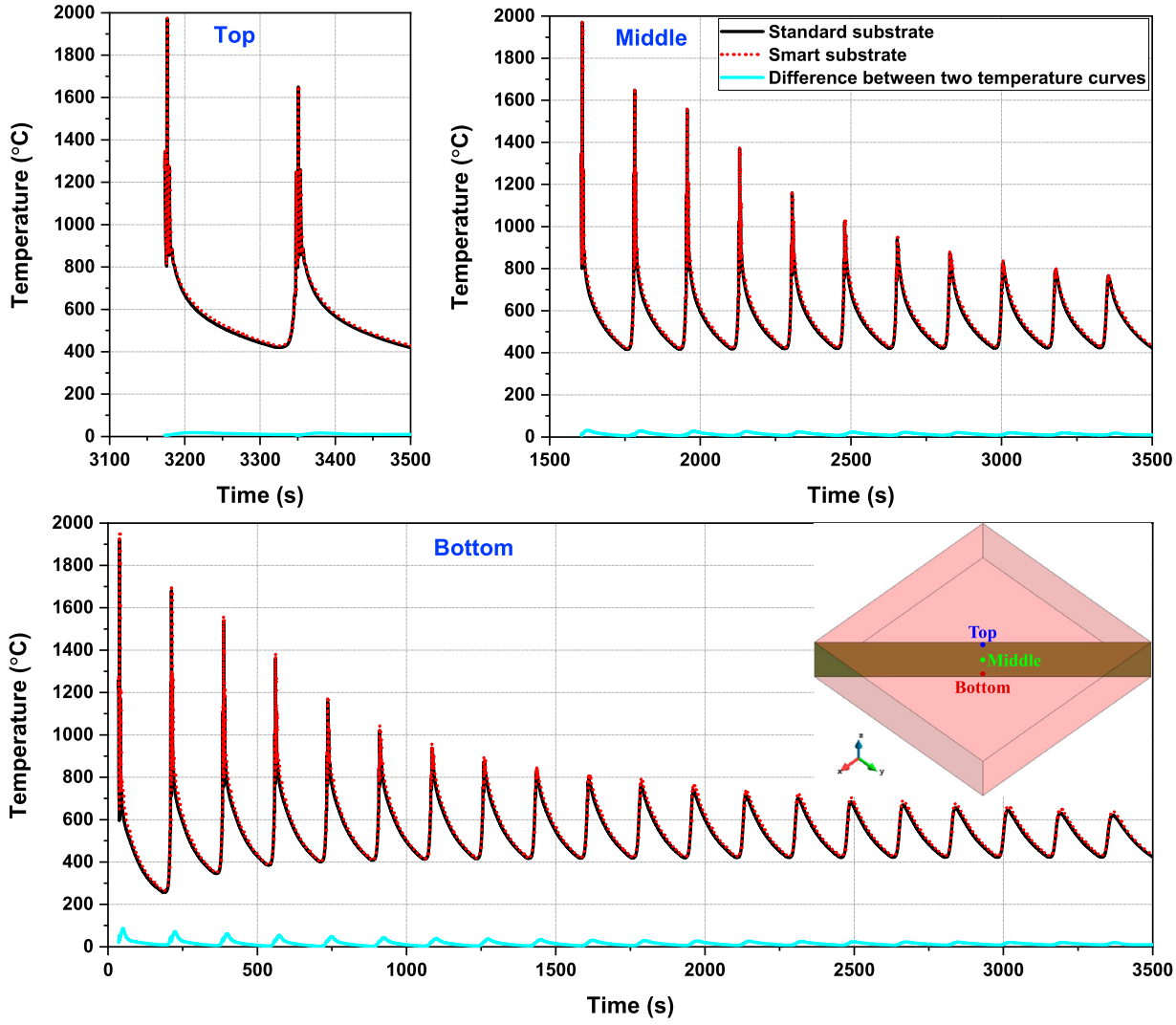


Figure 18. Comparison of the predicted temperature evolution at the building centre and different heights of the block parts fabricated on the standard and smart substrates.

avoid stress accretion during DED. The key is to design the proper moveable tenons (e.g. a clearance of 0.2-1.0 mm, depending on the material properties and the structural size) in the substrate to lower its strong mechanical constraining parallel to the scanning paths as it curbs the thermal deformation, generating residual stresses.

The basis of the Smart-Substrate approach is to avoid the accretion of residual stresses by allowing thermal strains to develop fully in the part. This allows a relatively larger shrinkage of the metal deposition, particularly in the horizontal direction, slightly lowering the geometrical precision of the printed components. This should be considered when designing the part using geometrical compensation.

While the application of Smart-Substrate barely affects the metallurgical microstructure and the mechanical hardness, its influence on other potential AM defects (e.g. surface roughness, lack of fusion, and porosities) and the fatigue performance of AM-components needs to be investigated in the future study, taking into account that fatigue life is closely related to these intrinsic defects, especially the residual stresses remaining in the builds [48].

5. Conclusions

In this study, an innovative structural design approach, called Smart-Substrate, is proposed to yield lower residual stresses and part deflections in DED process. The major findings are as follows:

(1) In AM, apart from MTG, the yield temperature, related to the constraint forces during AM thermal deformation, also plays a prominent role in the formation of residual stresses. Based on this, a novel

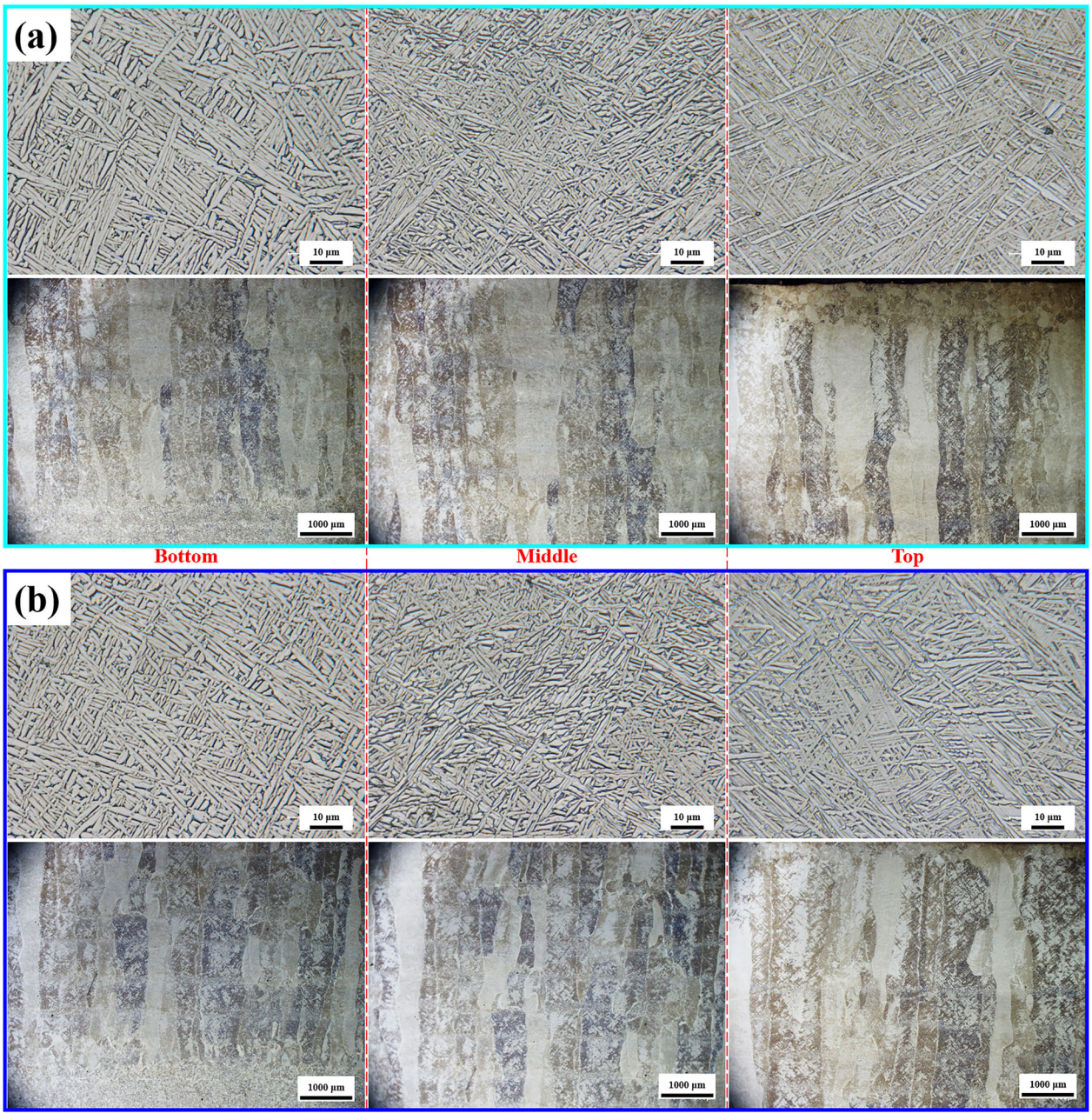


Figure 19. Comparison of the deposited macro/microstructures at the building centre and different heights of the block parts, corresponding to the thermal histories shown in Figure 17: (a) standard substrate; (b) Smart-Substrate.

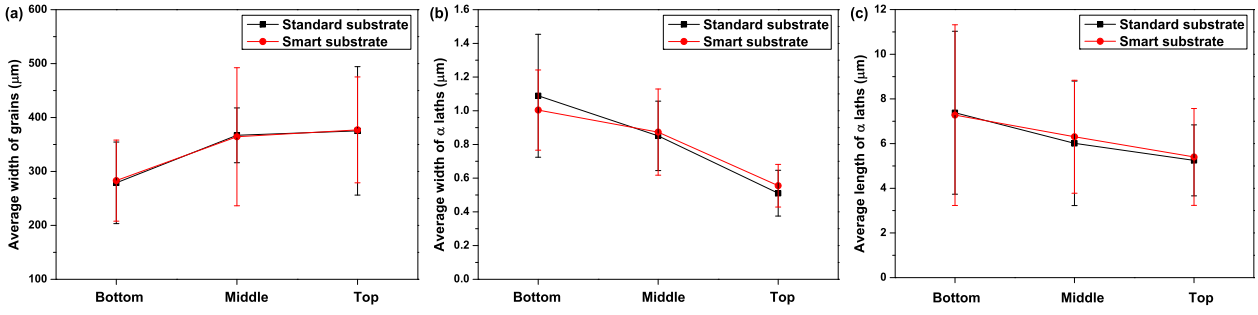


Figure 20. Comparison of the macro/microstructure size shown in Figure 19: (a) the average width of the β-grains; (b) the average width of the α laths; the average length of the α laths.

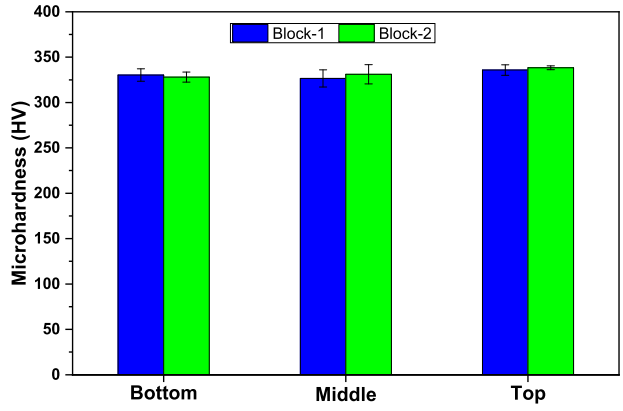


Figure 21. Comparison of the microhardness at the building centre and different heights of the two block parts.

Smart-Substrate strategy to avert stress accretion in DED is developed, different from MTG-based stress-control methods.

(2) Three groups of different DED structures with increasing geometrical complexity (single-wall, rectangular and block parts) are used to demonstrate the effectiveness of Smart-Substrate. It is found that using the standard substrate easily produces remarkable part deflections and large residual stresses, especially stress concentrations at the corner of the builds, but this is not the case for the Smart-Substrate samples.

(3) The key physical mechanism of applying the Smart-Substrate approach is to diminish the local mechanical stiffness of the upper part of the substrate to allow strains in the metal deposition to develop fully in an increased elastic range. Thus, the formation of residual stresses and plastic strains is curbed at the cost of slightly lowering the geometric precision of the build due to its freer volumetric contractions. This may be corrected in the design of the part by geometrical compensation.

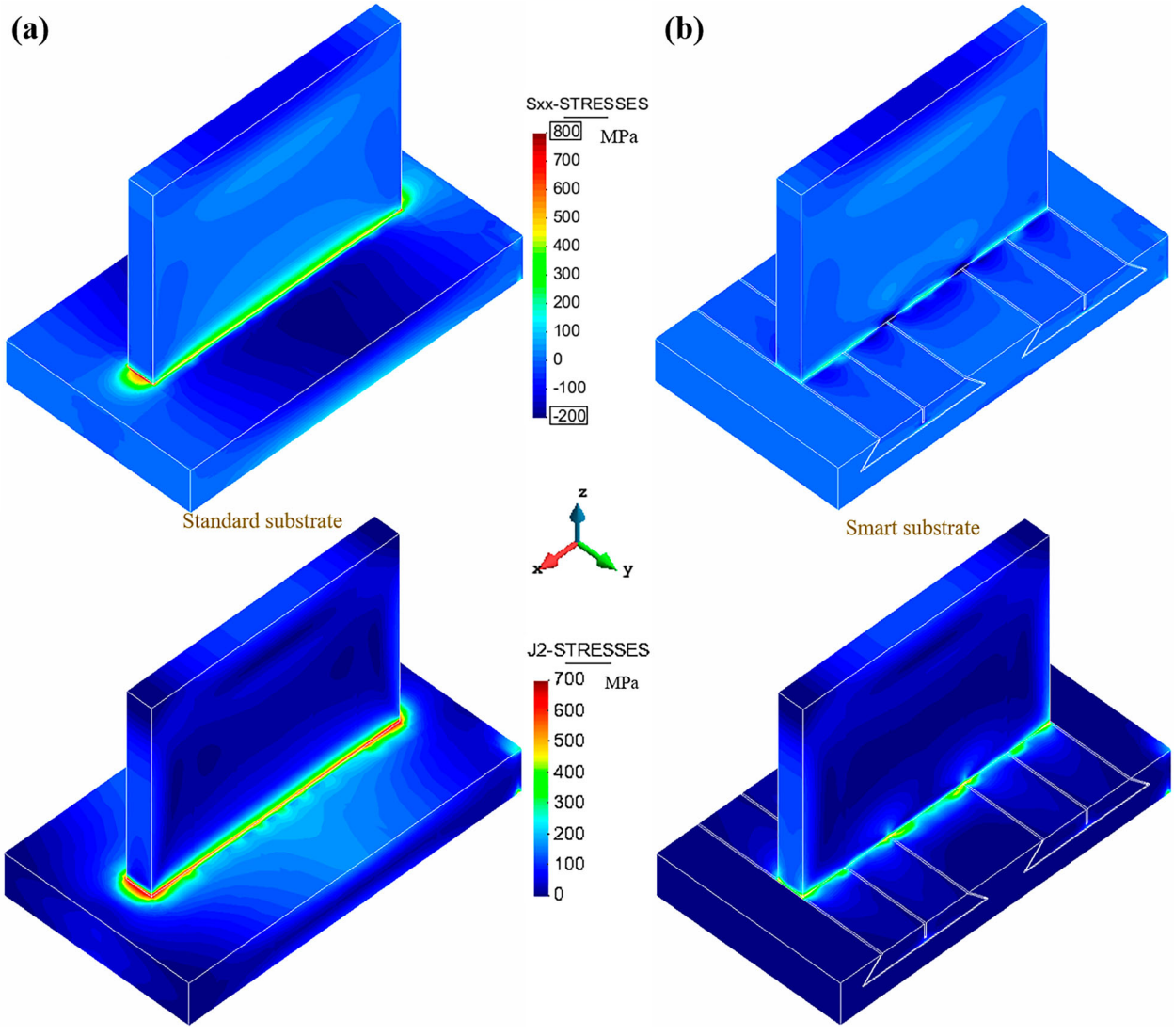


Figure 22. Residual longitudinal (S_{xx}) and von Mises stresses of two single-walls with an increased height of 50 mm built on (a) the standard and (b) the smart substrate.

(4) Unlike MTG-based strategies, the Smart-Substrate strategy shows an unimportant influence on the thermal and metallurgical evolution as well as the resulting mechanical hardness because it preserves an effective heat conduction channel. This provides the possibility to address the stress problems, the microstructure design and the defect prevention, separately, for obtaining superior mechanical properties of AM-parts.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

[1] DebRoy T, Wei H, Zuback J, et al. Additive manufacturing of metallic components–process, structure and properties. Prog Mater Sci. 2018;92:112–224. doi:10.1016/j.pmatsci.2017.10.001

[2] Gu D, Shi X, Poprawe R, et al. Material-structure-performance integrated laser-metal additive manufacturing. Science. 2021;372(6545):eabg1487. doi:10.1126/science.abg1487

[3] Frazier W. Metal additive manufacturing: a review. J Mater Eng Perform. 2014;23:1917–1928. doi:10.1007/s11665-014-0958-z

[4] Mugwagwa L, Yadroitsava I, Makoana N, et al. Residual stress in laser powder bed fusion. In: *Fundamentals of laser powder bed fusion of metals*. Elsevier; 2021. p. 245–276. doi:10.1016/B978-0-12-824090-8.00014-7

[5] Li R, Wang G, Zhao X, et al. Effect of path strategy on residual stress and distortion in laser and cold metal transfer hybrid additive manufacturing. Addit Manuf. 2021;46:102203. doi:10.1016/j.addma.2021.102203

[6] He P, Sun C, Wang Y. Material distortion in laser-based additive manufacturing of fuel cell component: three-dimensional numerical analysis. Addit Manuf. 2021;46:102188. doi:10.1016/j.addma.2021.102188

[7] Platl J, Bodner S, Hofer C, et al. Cracking mechanism in a laser powder bed fused cold-work tool steel: The role of residual stresses, microstructure and local elemental concentrations. Acta Mater. 2022;225:117570. doi:10.1016/j.actamat.2021.117570

[8] Magana-Carranza R, Sutcliffe C, Patterson E. The effect of processing parameters and material properties on residual forces induced in laser powder Bed fusion (L-PBF). Addit Manuf. 2021;46:102192. doi:10.1016/j.addma.2021.102192

[9] Sun J, Hensel J, Köhler M, et al. Residual stress in wire and arc additively manufactured aluminum components. J Manuf Process. 2021a;65:97–111. doi:10.1016/j.jmapro.2021.02.021

[10] Lu X, Lin X, Chiumenti M, et al. Finite element analysis and experimental validation of the thermomechanical behavior in laser solid forming of Ti-6Al-4V. Addit Manuf. 2018;21:30–40. doi:10.1016/j.addma.2018.02.003

[11] Wu Q, Mukherjee T, Liu C, et al. Residual stresses and distortion in the patterned printing of titanium and nickel alloys. Addit Manuf. 2019;29:100808. doi:10.1016/j.addma.2019.100808

[12] Xu F, Lv Y, Liu Y, et al. Microstructural evolution and mechanical properties of Inconel 625 alloy during pulsed plasma arc deposition process. J Mater Sci Technol. 2013;29(5):480–488. doi:10.1016/j.jmst.2013.02.010

[13] Gornyakov V, Ding J, Sun Y, et al. Understanding and designing post-build rolling for mitigation of residual stress and distortion in wire arc additively manufactured components. Mater Des. 2022;213:110335. doi:10.1016/j.matdes.2021.110335

[14] Tong Z, Ren X, Jiao J, et al. Laser additive manufacturing of FeCrCoMnNi high-entropy alloy: effect of heat treatment on microstructure, residual stress and mechanical property. J Alloys Compd. 2019;785:1144–1159. doi:10.1016/j.jallcom.2019.01.213

[15] Sun R, Li L, Zhu Y, et al. Microstructure, residual stress and tensile properties control of wire-arc additive manufactured 2319 aluminum alloy with laser shock peening. J Alloys Compd. 2018;747:255–265. doi:10.1016/j.jallcom.2018.02.353

[16] Zhang Q, Duan B, Zhang Z, et al. Effect of ultrasonic shot peening on microstructure evolution and corrosion resistance of selective laser melted Ti–6Al–4 V alloy. J Mater Res Technol. 2021;11:1090–1099. doi:10.1016/j.jmrt.2021.01.091

[17] Ueda Y, Murakawa H, Ma N. Chapter 1-introduction to welding mechanics. In: *Welding deformation and residual stress prevention*. Boston: Butterworth-Heinemann; 2012.. doi:10.1016/B978-0-12-394804-5.00001-3

[18] Jing H, Ge P, Zhang Z, et al. Numerical studies of the effects of the substrate structure on the residual stress in laser directed energy additive manufacturing of thin-walled products. Metals (Basel). 2022;12(3):462. doi:10.3390/met12030462

[19] Lu X, Chiumenti M, Cervera M, et al. Substrate design to minimize residual stresses in directed energy deposition AM processes. Mater Des. 2021;202:109525. doi:10.1016/j.matdes.2021.109525

[20] Wang J, Lin X, Li J, et al. Effects of deposition strategies on macro/microstructure and mechanical properties of wire and arc additive manufactured Ti6Al4 V. Mater Sci Eng A. 2019;754:735–749. doi:10.1016/j.msea.2019.03.001

[21] Wang Z, Lin X, Kang N, et al. Directed energy deposition additive manufacturing of a Sc/Zr-modified Al–Mg alloy: effect of thermal history on microstructural evolution and mechanical properties. Mater Sci Eng A. 2021;802:140606. doi:10.1016/j.msea.2020.140606

[22] Fang Z, Wu Z, Huang C, et al. Review on residual stress in selective laser melting additive manufacturing of alloy parts. Opt Laser Technol. 2020;129:106283. doi:10.1016/j.optlastec.2020.106283

[23] Tran H, Liang X, To A. Efficient prediction of cracking at solid-lattice support interface during laser powder bed

fusion via global-local J-integral analysis based on modified inherent strain method and lattice support homogenization. Addit Manuf. 2020;36:101590. doi:10.1016/j.addma.2020.101590

[24] Song X, Feih S, Zhai W, et al. Advances in additive manufacturing process simulation: residual stresses and distortion predictions in complex metallic components. Mater Des. 2020;193:108779. doi:10.1016/j.matdes.2020.108779

[25] Khoo Z, Teoh J, Liu Y, et al. 3D printing of smart materials: A review on recent progresses in 4D printing. Virtual Phys Prototyp. 2015;10(3):103–122. doi:10.1080/17452759.2015.1097054

[26] Lee A, An J, Chua C. Two-way 4D printing: a review on the reversibility of 3D-printed shape memory materials. Engineering. 2017;3(5):663–674. doi:10.1016/J.ENG.2017.05.014

[27] Chiumenti M, Cervera M, Salmi A, et al. Finite element modeling of multi-pass welding and shaped metal deposition processes. Comput Methods Appl Mech Eng. 2010;199(37-40):2343–2359. doi:10.1016/j.cma.2010.02.018

[28] Lu X, Lin X, Chiumenti M, et al. In situ measurements and ThermoMechanical simulation of Ti-6Al-4V laser solid forming processes. Int J Mech Sci. 2019;119-130:119–130. doi:10.1016/j.ijmecsci.2019.01.043

[29] Chiumenti M, Neiva E, Salsi E, et al. Numerical modelling and experimental validation in selective laser melting. Addit Manuf. 2017;18:171–185. doi:10.1016/j.addma.2017.09.002

[30] Li Y, Zhou K, Tan P, et al. Modeling temperature and residual stress fields in selective laser melting. Int J Mech Sci. 2018;136:24–35. doi:10.1016/j.ijmecsci.2017.12.001

[31] Denlinger E, Michaleris P. Effect of stress relaxation on distortion in additive manufacturing process modeling. Addit Manuf. 2016;12:51–59. doi:10.1016/j.addma.2016.06.011

[32] Chen Z, Ye H, Xu H. Distortion control in a wire-fed electron-beam thin-walled Ti-6Al-4V freeform. J Mater Process Technol. 2018;258:286–295. doi:10.1016/j.jmatprotec.2018.04.008

[33] Chiumenti M, Cervera M, Dialami N, et al. Numerical modeling of the electron beam welding and its experimental validation. Finite Elem Anal Des. 2016;121:118–133. doi:10.1016/j.finel.2016.07.003

[34] Lu X, Chiumenti M, Cervera M, et al. Mitigation of residual stresses and microstructure homogenization in directed energy deposition processes. Eng Comput. 2022: 4771–4790. doi:10.1007/s00366-021-01563-9

[35] Cao J, Gharghouri M, Nash P. Finite-element analysis and experimental validation of thermal residual stress and distortion in electron beam additive manufactured Ti-6Al-4V build plates. J Mater Process Technol. 2016;237:409–419. doi:10.1016/j.jmatprotec.2016.06.032

[36] Fallah V, Alimardani M, Corbin S, et al. Temporal development of melt-pool morphology and clad geometry in laser powder deposition. Comput Mater Sci. 2011;50(7):2124–2134. doi:10.1016/j.commatsci.2011.02.018

[37] Promoppatum P, Yao S. Influence of scanning length and energy input on residual stress reduction in metal additive manufacturing: numerical and experimental studies. J Manuf Process. 2020;49:247–259. doi:10.1016/j.jmapro.2019.11.020

[38] Ramos D, Belblidia F, Sienz J. New scanning strategy to reduce warpage in additive manufacturing. Addit Manuf. 2019;28:554–564. doi:10.1016/j.addma.2019.05.016

[39] Ivanov S, Artinov A, Zemlyakov E, et al. Spatiotemporal evolution of stress field during direct laser deposition of multilayer thin wall of Ti-6Al-4V. Materials (Basel). 2022;15(1):263. doi:10.3390/ma15010263

[40] Nycz A, Lee Y, Noakes M, et al. Effective residual stress prediction validated with neutron diffraction method for metal large-scale additive manufacturing. Mater Des. 2021;205:109751. doi:10.1016/j.matdes.2021.109751

[41] Sun L, Ren X, He J, et al. Numerical investigation of a novel pattern for reducing residual stress in metal additive manufacturing. J Mater Sci Technol. 2021b;67:11–22. doi:10.1016/j.jmst.2020.05.080

[42] Yang Q, Zhang P, Cheng L, et al. Finite element modeling and validation of thermomechanical behavior of Ti-6Al-4V in directed energy deposition additive manufacturing. Addit Manuf. 2016;12:169–177. doi:10.1016/j.addma.2016.06.012

[43] Chen C, Xiao Z, Zhang W, et al. Effect of laser jump speed on temperature distribution and thermal stress in laser powder bed fusion. Opt Laser Technol. 2021;142:107275. doi:10.1016/j.optlastec.2021.107275

[44] Jiang X, Ye T, Zhu Y. Effect of process parameters on residual stress in selective laser melting of AlSi10Mg. Mater Sci Technol. 2020;36(3):342–352. doi:10.1080/02670836.2019.1705560

[45] Xiao Z, Chen C, Zhu H, et al. Study of residual stress in selective laser melting of Ti6Al4 V. Mater Des. 2020;193:108846. doi:10.1016/j.matdes.2020.108846

[46] Carraturo M, Kollmannsberger S, Reali A, et al. An immersed boundary approach for residual stress evaluation in selective laser melting processes. Addit Manufact. 2021;46:102077. doi:10.1016/j.addma.2021.102077

[47] Szost B, Terzi S, Martina F, et al. A comparative study of additive manufacturing techniques: residual stress and microstructural analysis of CLAD and WAAM printed Ti–6Al–4 V components. Mater Des. 2016;89:559–567. doi:10.1016/j.matdes.2015.09.115

[48] Sanaei N, Fatemi A. Defects in additive manufactured metals and their effect on fatigue performance: a state-of-the-art review. Prog Mater Sci. 2021;117:100724. doi:10.1016/j.pmatsci.2020.100724